Mobile Immune Signals as a Body-Brain Communication Network:

Typed Causal Routes, Receiver-Relative Navigation, and a Cellular Oscillating Tomography Research Program

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Abstract

A cut finger creates two linked coordination problems. Mobile immune cells must navigate a changing wound, while nervous and immune systems must exchange information so that repair, inflammation, behavior, pain, energy use, and later readiness remain coordinated. Generic labels such as "body–brain axis" conceal the distinct senders, carriers, receivers, transformations, delays, and endpoints that make these routes testable.

This paper develops a typed causal architecture for body–brain communication and immune-cell navigation. Established chemical gradients, immune relay waves, neural and neuropeptide modulation, vagal cytokine sensing, brain-border interfaces, vascular transformation, and returned physiology remain explicit. The proposed Self-Aware Networks extension is narrower: after those established variables are measured, does a receiver-relative residual defined against a cell's receptor, timing, and state history add held-out directional information? Phase or spectral variables enter only if the corresponding signal is physically measurable and reliable.

The paper specifies two staged biological experiments and reports a deterministic synthetic reference application. Under generator shift, the preserved classifier passed the complete gate in only 3 of 10 non-mixture families; all 24 adverse, null, and refusal rows and all mixture refusals remain part of the result. A later development-only origin-set estimator recovered 344 of 400 single-origin cases (accuracy 0.860; 95% Wilson interval 0.823–0.891) and 376 of 470 complete origin sets (0.800; 0.761–0.834). Mixture exactness was lower at 46 of 70 (0.657; 0.540–0.758), although none of the 70 mixtures was compressed into a singleton. A fresh replay reproduced all twelve scientific outputs byte-for-byte. Fourteen Lean theorems machine-check finite refusal, compatibility, and accounting rules with no admitted proofs; they do not establish any biological premise.

The application therefore demonstrates an inspectable method for issuing, withholding, and auditing route certificates under declared synthetic mechanisms. It does not establish neural steering of immune cells, a phase-bearing immune signal, a clinical effect, or biological validity. Those questions remain assigned to the staged intervention-and-rescue program.

Keywords: neuroimmune communication; T cell; neutrophil; chemotaxis; vagus nerve; cytokine; wound healing; Cellular Oscillating Tomography; Self-Aware Networks; phase-wave differential; receiver-relative computation

Reader contract: the vocabulary must be earned

The exposition begins with an ordinary event, identifies the unexplained scientific problem, describes established routes, and states the proposed operation in plain language before naming SAN terminology. A coined term is useful only if a reader can remove it and still say what receives information, what changes, what is transmitted next, and what measurement would distinguish the proposal from an established alternative. Cellular Oscillating Tomography and phase-wave differential therefore summarize already-described candidate operations; they do not replace cell class, carrier, anatomy, intervention, timing, or endpoint.

1 The ordinary problem: finding a wound while the organism is changing

Imagine a neutrophil or T cell moving through tissue after an injury. It does not receive a street address. It encounters concentrations that vary across space, ligands that bind and unbind, adhesion sites, mechanical barriers, oxygen and metabolic changes, signals released by damaged cells, signals relayed by other immune cells, and neuropeptides released from nearby nerve endings. The receiver itself is changing as receptors adapt, intracellular pathways activate, metabolism shifts, and the cytoskeleton builds a new leading edge.

At the same time, the organism changes behavior. Pain protects the wound. Sickness can reduce exploration and redirect energy. Neural circuits sense inflammatory mediators and alter peripheral immune output. Later, repair signals suppress recruitment, increase clearance, and help end the response. The body therefore faces a coupled estimation-and-control problem:

> *How can moving cells estimate what direction to take while body and brain continuously alter the signals being estimated?*

The question becomes harder when one asks what "communication" means. A cytokine binding a vagal sensory neuron, a vagal action potential reaching the brainstem, acetylcholine released by a T cell in the spleen, IL-4 acting on a GABAergic neuron, CGRP acting on a wound macrophage, and a neutrophil following an LTB4 relay are all genuine communication events. They are not one physical channel. Their senders, receivers, velocities, transformations, noise, and biological consequences differ.

The paper begins from those differences. A joined body–brain network is scientifically useful only when its route types remain visible.

2 What a complete explanation must separate

Six operations are frequently blended together.

1. **Local source formation.** Damage, pathogen products, dying cells, vascular change, and tissue mechanics create local chemical and physical fields.
2. **Cellular navigation.** A moving cell samples those fields through receptors and converts asymmetric evidence into polarization, adhesion, and motion.
3. **Collective relay.** Responding immune cells amplify, propagate, limit, or terminate recruitment signals.

4. **Neural-to-immune modulation.** Sensory or autonomic neurons release transmitters or neuropeptides that alter immune-cell recruitment, secretion, survival, or phenotype.
5. **Immune-to-neural transmission.** Cytokines, immune cells, barriers, vagal afferents, meninges, glia, and neuronal receptors change neural state.
6. **Closed-loop regulation.** Brain circuits return signals that alter inflammation, behavior, energy allocation, and the next body-derived input.

The first three can occur without the brain. The fourth and fifth can be local or systemic. The sixth requires a recurrent route whose direction changes across the loop. A model that observes a behavioral endpoint but omits these distinctions cannot identify which operation failed.

The same discipline applies to cell identity. Neutrophils are rapid innate responders with well-studied swarming dynamics. Conventional αβ T cells, γδ T cells, macrophages, monocytes, microglia, B cells, and meningeal or parenchymal immune populations have different programs. Evidence that a neutrophil follows an LTB4 relay cannot be relabeled as evidence that a T cell follows a neural address. It can, however, establish a reusable operation: a moving immune receiver can extract directional information from a signal generated by other cells and can participate in a self-limiting relay.

3 Established components of the network

3.1 Receptor sampling and the physical limits of direction sensing

Berg and Purcell showed that chemical sensing is constrained by diffusion, receptor occupancy, integration time, and stochastic arrival [1]. Endres and Wingreen extended the quantitative treatment to direct spatial gradient sensing [2]. These results matter because a cell does not read an exact concentration field. It estimates one from finite, noisy receptor events.

For a receptor-bearing cell of effective size a, integration time T, ligand diffusivity D, and local concentration c, the fractional uncertainty has a lower-bound scaling of the form

$$\frac{\mathrm{Var}(\widehat{c})}{c^2} \gtrsim \frac{K}{DacT}, \tag{1}$$

where K depends on geometry, receptor kinetics, and the estimator. The inequality is not a universal numerical law for every immune cell. It states the relevant design constraint: longer sampling, more accessible ligand, favorable geometry, and integration across receptors can improve an estimate, while fast motion and fluctuating fields can make it worse.

This immediately motivates receiver-relative computation. Two cells at the same location need not infer the same direction because their receptor composition, adaptation state, recent history, and internal noise differ.

3.2 Wound gradients and immune relay waves

In zebrafish, Katikaneni and colleagues identified a 5-lipoxygenase-dependent lipid-peroxidation process supporting long-range wound detection and leukocyte attraction [3]. In human neutrophils, Strickland and colleagues measured calcium-linked recruitment waves in which LTB4 helps propagate swarming and self-termination prevents uncontrolled spread [4]. Together these studies replace the picture of a single static attractant with a dynamic field shaped by source, relay, and termination.

Let $c_k(\mathbf{x}, t)$ denote the concentration of wound-associated field k. A minimal field model is

$$\frac{\partial c_k}{\partial t} = D_k \nabla^2 c_k - \lambda_k c_k + S_k(\mathbf{x}, t), \tag{2}$$

where D_k is diffusion, λ_k is removal or decay, and S_k is the time-varying tissue source. A relay field r adds mobile sources:

$$\frac{\partial r}{\partial t} = D_r \nabla^2 r - \lambda_r r + \sum_j q_j(t)\, \delta(\mathbf{x} - \mathbf{X}_j(t)), \tag{3}$$

where $q_j(t)$ is signal release by responding cell j. Adaptation or inhibitory feedback can reduce q_j, producing a wave that recruits without expanding indefinitely.

The distinction between equations (2) and (3) is experimentally important. Blocking the wound source and blocking immune relay should have different spatial and temporal consequences.

3.3 Neural control at injured and inflamed tissue

Neural influence is not uniformly excitatory or attractive. In mouse rosacea-like dermatitis, sensory-neuron CGRP recruited and activated RAMP1-expressing γδ T cells and promoted IL-17A-related inflammation [13]. In acute mouse skin and muscle injury, Lu and colleagues found that nociceptor endings grew into injured tissue and used CGRP to regulate neutrophils, monocytes, and macrophages [14]. CGRP inhibited recruitment in tested assays, increased death under inflammatory conditions, promoted efferocytosis, and shifted macrophages toward repair.

These results supply a more precise operation than “neurons guide immune cells.” Neural signals can change the gain, sign, phase, and stopping conditions of an immune response. The same carrier can have different consequences because receptor expression, inflammatory phase, and receiver state differ.

3.4 Cytokine sensing and a body–brain reflex

The inflammatory reflex provides an explicit neural–immune relay. Rosas-Ballina and colleagues identified acetylcholine-synthesizing T cells required for vagus-nerve-mediated inhibition of cytokine production in mice [5]. Jin and colleagues later resolved an ascending component: pro- and anti-inflammatory cytokines activated distinguishable vagal sensory populations, which signaled to brainstem neurons; selective silencing dysregulated inflammation, whereas activation biased the response toward an anti-inflammatory state [11].

Huerta and colleagues then recorded individual nodose ganglion neurons during cytokine exposure [12]. Some neurons responded selectively to individual cytokines, while others responded to multiple cytokines with distinguishable activity patterns. Experimental colitis raised baseline activity while reducing responsiveness to selected cytokines. The receiver’s context therefore changed the neural representation of the same class of input.

This is direct evidence for a principle central to the present model: biological messages are receiver-state dependent. A cytokine is not a complete instruction by itself. Its effect depends on receptors, baseline activity, history, competing signals, and downstream connectivity.

3.5 Immune signals alter neural plasticity, memory, and behavior

Meningeal and brain-associated immune routes demonstrate the reverse direction. Derecki and colleagues linked meningeal T-cell-derived IL-4 to meningeal myeloid state and learning in mice [6]. Ribeiro and colleagues found that meningeal γδ T-cell-derived IL-17 affected glial BDNF, hippocampal plasticity, and short-term memory [7]. Alves de Lima and colleagues linked meningeal γδ T-cell IL-17A to neuronal receptors and anxiety-like behavior [8]. Herz and colleagues showed that GABAergic neuronal IL-4 receptors mediated a T-cell effect on memory [9].

Koren and colleagues supplied a further loop: mouse insular-cortex ensembles active during particular inflammatory states could later be reactivated to retrieve aspects of those inflammatory responses [10]. The result supports a neural representation of immune history without implying that one cortical ensemble contains every molecular detail of the body's immune state.

Finally, recent anatomical work expands the mobile-cell component. Yoshida and colleagues identified extravascular αβ T cells concentrated in the subfornical organ in mice and humans; in mice, gut- and adipose-associated histories contributed to the population, CXCR6 supported retention, IFNγ contributed to homeostasis, and manipulations affected food-seeking behavior [15]. Smolders and colleagues independently characterized tissue-resident-memory-like T cells in human brain tissue [16]. These studies establish mobile and resident immune participation in brain-associated compartments while leaving exact homing histories and information bandwidth as open questions.

3.6 Route taxonomy and endpoint discipline

The route taxonomy makes the route types explicit before applying the SAN extension:

The same biological molecule may occur in multiple route classes, but its role is preparation- and receiver-dependent. A paper-level comparison must therefore name the carrier, cell class, transformation, delay, direction, tissue, and endpoint for each claimed edge. "Neuroimmune communication" is a family name, not one mechanism.

3.7 Three newer routes sharpen the boundary

Three recent primary studies broaden the evidence without collapsing the route types. In a female-derived human in-vitro neurogenesis model, Nissen and colleagues found that chronic TNF-α exposure induced a type-I-interferon loop in hippocampal progenitor cells, promoted chemokine-mediated, CXCR3-dependent T-cell recruitment, and suppressed neurogenesis [24]. This is a directly measured cytokine-to-progenitor-to-chemokine-to-T-cell route in a bounded culture system. It is not evidence that the same route operates with the same magnitude in an intact human hippocampus.

In mouse lung adenocarcinoma models, Wei and colleagues identified an afferent–efferent loop: tumour-associated signals engaged Npy2r-expressing vagal sensory neurons, brainstem activity increased sympathetic output, and β2-adrenergic signaling in alveolar macrophages suppressed anti-tumour immunity [25]. Genetic, pharmacological, and chemogenetic interruptions changed downstream immunity and tumour

Route class	Sender and carrier	Receiver	Measured transformation	Primary endpoint	What it does not establish
Chemical gradient	Wound tissue and soluble lipid/cytokine fields	Motile leukocyte receptors	Diffusion, degradation, receptor sampling, polarization	Directional bias or recruitment	A neural address or point-to-point message
Immune relay	Responding neutrophils or other immune cells and LTB4-linked release	Neighboring immune cells	Release, amplification, self-termination	Swarm recruitment and containment	T-cell navigation or brain control
Neural-to-immune	Sensory neuron, vagal neuron, CGRP, acetylcholine	Immune cell or inflammatory circuit	Receptor-specific gain/sign change and reflex control	Cytokine output, recruitment, repair, or inflammation	Universal attraction or a semantic code
Immune-to-neural	Cytokine, T-cell signal, or meningeal immune factor	Vagal, brainstem, cortical, or GABAergic receiver	Receptor activation and network-state change	Plasticity, memory, affect, or inflammatory-state representation	A migrating cell carrying a complete neural scene
Cellular migration/retention	Mobile or resident T-cell population	Tissue niche such as SFO or brain-associated compartment	Chemokine retention, adhesion, and tissue entry	Localization, persistence, and behavioral association	Neuron-directed point navigation
Receiver-relative residual	Any measured carrier after baseline conditioning	A specified receiving cell or circuit	Deviation from receiver-conditioned expectation	Incremental held-out trajectory prediction and intervention loss	A PWD merely because a signal is oscillatory

growth. This establishes that a recurrent body–brain–body path can be decomposed and perturbed edge by edge. It does not establish wound-cell steering or the receiver-relative residual proposed here.

The RESET-RA trial provides a separate human boundary. In 242 people with rheumatoid arthritis, active vagus-nerve stimulation produced a higher three-month ACR20 response than sham stimulation, with the reported primary endpoint met [26]. That result is clinically important evidence that a neural intervention can alter an immune-associated disease endpoint. It does not identify every biological mediator, validate the mobile-cell navigation model, or justify self-treatment. The device, implantation, patient selection, adverse-event monitoring, and clinical endpoint belong to a regulated medical context.

4 The source-faithful development of the SAN problem

The present model comes from a staged Self-Aware Networks genealogy. The stages should be connected without making the later vocabulary appear earlier than it did.

4.1 A coupled fast–slow cellular state in 2017

In a May 2017 public Neural Lace Podcast discussion, I asked what a hypothetical thirty-second snapshot of every neuron and astrocyte would capture. The conversation distinguished fast ionotropic activity from slower metabotropic, receptor-mediated, hormonal, and chemical state, and described a neuronal network and glial network as interacting systems. The point was not that one omitted variable explains consciousness. It was that a neuron-only instantaneous state can be causally incomplete when slower cellular conditions determine what fast activity does next [17].

That architecture is an ancestor of the current body–brain model. It does not yet name T-cell guidance or the later COT and phase-wave-differential terms.

4.2 The mobile-cell guidance problem in 2021 and 2022

A 2021 note used a concrete biological puzzle: ordinary cells recognize conditions and coordinate other cells over distance, exemplified by point-by-point guidance of a white blood cell toward an injury [18]. A March 2022 recorded discussion expanded the problem into a moving target–relay–traveler system. The wound produced chemical, mechanical, thermal, and oscillatory consequences; responding cells contributed further signals; a mobile white blood cell changed course; and the discussion asked whether neural activity joined the recurrent process [19].

The complete context matters. The recording explicitly left the white-cell theory unfinished and then moved into the broader question of how biological cells exceed simplified artificial-neuron models. It is therefore a source for the problem architecture, not evidence that a neural dispatcher had already been demonstrated.

4.3 The neuron–T-cell and COT synthesis in 2022

Public SAN notes in June 2022 joined gradients, neutrophil swarming, receptor state, cytokine and neurotransmitter exchange, mobile T cells, immune memory, neural effects, and information-theoretic cellular communication [20]. A separate versioned note introduced Cellular Oscillating Tomography (COT) as an attempt to describe how cells receive patterns through receptors, transform them through current and learned intracellular state, and transmit new consequences. It explicitly placed T-cell and glial interfaces inside the broader COT-to-NAPOT scope [21].

The strongest reading is operational. Cells of different classes can participate in recurrent receive–transform–transmit loops while using different physical carriers. `Wireless` in the historical note is best translated as mobile-cell and soluble-signal relay. The scientific proposition does not require a T cell to emit radio waves or an immune cell to carry a duplicate sensory image.

4.4 Receiver-relative addressing in 2025

In a February 2025 working dialogue, I sharpened the earlier guidance problem. If every cell simply emitted an absolute frequency "address," the moving cell would not need both the target and a relay. The proposed alternative was relational: the injury supplies an originating pattern, another cell or neural route supplies a transformed signal, and the moving receiver updates its direction from the changing relation between those inputs and its own expected state [22].

This stage introduced phase-wave-differential, coefficient-of-variation, and spectral language into the navigation question. It is a later extension, not hidden terminology inside the 2021 note or 2022 recording.

5 The operation before the terminology

The joined operation can now be stated without an acronym.

1. A wound changes local chemistry, mechanics, temperature, vascular state, and cellular activity.
2. A mobile immune cell samples a small, noisy portion of those changes through its receptors.
3. Nearby responding cells may amplify or terminate a relay.

4. Sensory or autonomic neurons may detect inflammatory variables and release signals that change immune-cell readiness, migration, survival, or phenotype.
5. The moving cell compares new samples with its recent history and internal state, polarizes, and moves.
6. Its new position changes what it samples next.
7. In parallel, immune variables are encoded by sensory neural routes, transformed in brain circuits, and returned as regulation of immune state and behavior.

Only after that causal chain is visible is **Cellular Oscillating Tomography** useful as a compact name for the receiver-centered operation: a cell samples distributed signals, transforms them through its current and learned state, and emits an updated consequence that becomes input to other receivers.

The navigation extension introduces a second name. A **receiver-relative phase-wave differential** is the measured departure of a current input relation from the timing or state relation expected by that receiver. It is not a synonym for chemotaxis, phase coding, calcium oscillation, or coefficient of variation. It is a candidate additional predictor that must earn its place against those established variables.

6 A typed formal model

6.1 State and route types

Let the measured organismal state be

$$\mathbf{s}(t) = \big[\mathbf{c}(t), \mathbf{r}(t), \mathbf{n}(t), \mathbf{z}(t), \mathbf{X}(t), \mathbf{b}(t)\big], \tag{4}$$

where $\mathbf{c}$ contains wound and cytokine fields, $\mathbf{r}$ immune relay fields, $\mathbf{n}$ neural or neuropeptide measurements, $\mathbf{z}$ receiver-cell states, $\mathbf{X}$ cell positions, and $\mathbf{b}$ brain and behavioral state.

A route edge is typed as

$$e = (u, \kappa, T, v, \tau, \eta), \tag{5}$$

with sender u, carrier class κ, transformation T, receiver v, delay τ, and uncertainty or noise model η. Cytokine diffusion, vagal spikes, CGRP release, T-cell acetylcholine, and cell migration therefore occupy different edges even when they belong to one recurrent path.

A candidate path $P = e_1 \circ \cdots \circ e_m$ is supported only when perturbing an upstream edge changes the downstream mediator and endpoint with the predicted delay, and restoring the mediator downstream selectively restores the endpoint.

6.2 Receiver state

For mobile cell i, receptor and intracellular state evolve as

$$\mathbf{z}_i(t + \Delta t) = F_i(\mathbf{z}_i(t), \mathbf{y}_i(t), \mathbf{m}_i(t), \boldsymbol{\xi}_i(t)) \,, \tag{6}$$

where $\mathbf{y}_i$ is receptor input, $\mathbf{m}_i$ metabolic and mechanical context, and $\boldsymbol{\xi}_i$ unmeasured variation. This state changes receptor gain, polarization, secretion, adhesion, and the interpretation of later inputs.

A conventional gradient-and-relay motion model is

$$\dot{\mathbf{X}}_i(t) = v_i(t)\,\mathcal{N}\left[\sum_k \alpha_{ik}\nabla\log(c_k+\epsilon) + \beta_i\nabla\log(r+\epsilon) + \gamma_i\nabla n\right] + \boldsymbol{\omega}_i(t), \tag{7}$$

where $\mathcal{N}$ normalizes direction, coefficients may depend on $\mathbf{z}_i$, and $\boldsymbol{\omega}_i$ captures motility noise. Equation (7) is the baseline, not the SAN extension.

6.3 Receiver-relative residual

Let $\widehat{\boldsymbol{\mu}}_i(t)$ and $\widehat{\boldsymbol{\Sigma}}_i(t)$ be the receiver's expected feature mean and covariance given recent history and state. The standardized departure is

$$\mathbf{d}_i(t) = \widehat{\boldsymbol{\Sigma}}_i(t)^{-1/2}\left[\mathbf{f}_i(t) - \widehat{\boldsymbol{\mu}}_i(t)\right], \tag{8}$$

where $\mathbf{f}_i$ can include receptor occupancy, calcium state, cytokine timing, relay timing, neural or neuropeptide activity, and motion history. A coefficient of variation can be one component of $\mathbf{f}_i$; it is not the whole residual.

If phase can be measured meaningfully in the selected biological variables, a bounded relational feature is

$$\rho_i(t) = 1 - \cos\left(\phi_{W,i}(t) - \phi_{R,i}(t) - \theta_i(t)\right), \tag{9}$$

where $\phi_{W,i}$ is the phase of a wound-originated feature at the receiver, $\phi_{R,i}$ the transformed relay feature, and θ_i the receiver's learned or adaptive offset. Equation (9) has scientific meaning only when the underlying signals have a reproducible oscillatory component and phase estimation exceeds the noise floor.

The extended motion model is

$$\dot{\mathbf{X}}_i(t) = \dot{\mathbf{X}}_i^{(0)}(t) + \mathbf{W}_{d,i}\mathbf{d}_i(t) + \mathbf{w}_{\rho,i}\rho_i(t), \tag{10}$$

where $\dot{\mathbf{X}}_i^{(0)}$ is the established baseline in equation (7). The added terms are retained only if they improve held-out prediction and disappear under the predicted intervention.

6.4 Nested models and evidentiary threshold

The analysis compares four models:

$$\begin{aligned}
M_0 &: \text{wound fields and cell state},\\
M_1 &: M_0 + \text{immune relay},\\
M_2 &: M_1 + \text{neural or neuropeptide variables},\\
M_3 &: M_2 + \text{receiver-relative residual and, if measurable, phase relation}.
\end{aligned} \tag{11}$$

For held-out trajectory segments $\mathcal{H}$, model value is measured by predictive log score and angular error:

$$\Delta\mathcal{L}_{3,2} = \sum_{t\in\mathcal{H}} \log \frac{p(\Delta\mathbf{X}_i(t) \mid M_3)}{p(\Delta\mathbf{X}_i(t) \mid M_2)}, \qquad E_\angle = \frac{1}{|\mathcal{H}|}\sum_{t\in\mathcal{H}} \angle\left(\widehat{\Delta\mathbf{X}}_i, \Delta\mathbf{X}_i\right). \tag{12}$$

The SAN navigation term earns support only when $\Delta\mathcal{L}_{3,2} > 0$ across preregistered held-out animals or preparations, angular error falls, calibration remains valid, and route-specific perturbation removes the advantage.

7 Experiment A: identify a complete body–brain immune route

7.1 Preparation and measurements

Use a controlled peripheral inflammatory challenge in mice with simultaneous measurement of:

- circulating and local pro- and anti-inflammatory cytokines;
- activity in genetically defined vagal sensory populations;
- activity in the relevant nucleus tractus solitarius population;
- candidate descending autonomic or humoral mediators;
- T-cell, myeloid, and tissue response states;
- locomotion, temperature, feeding, pain-related behavior, corticosterone, and general sickness.

The goal is not to reproduce every existing result in one experiment. It is to test whether a typed route predicts the next state better than a generic "inflammation level" variable.

7.2 Factorial perturbation and rescue

Interventions are organized by edge:

1. block a selected cytokine or its receptor on the relevant vagal population;
2. silence the vagal afferent population;
3. silence the identified brainstem population;
4. interrupt the candidate descending effector;
5. restore the predicted mediator downstream of the interrupted edge.

For mediator m, endpoint Y, and intervention A, define the matched-rescue fraction

$$R_m = \frac{\mathbb{E}[Y \mid do(A=0, m=m^*)] - \mathbb{E}[Y \mid do(A=0)]}{\mathbb{E}[Y \mid do(A=1)] - \mathbb{E}[Y \mid do(A=0)]}. \tag{13}$$

A value near one indicates restoration of the measured endpoint under the specified conditions; it does not by itself prove that m is the only natural mediator. Rescue must also be selective: locomotion or corticosterone rescue without restoration of the typed immune consequence is not route completion.

7.3 Competing explanations

The preregistered alternatives include generic inflammation, global arousal, sickness behavior, stress hormone change, pain, temperature, and nonspecific viral or chemogenetic effects. These variables are measured rather than discussed after the fact. Cell-type specificity, temporal order, dose response, and matched sham interventions are required.

8 Experiment B: test receiver-relative immune-cell navigation

8.1 Stage the question by cell class

The fastest initial preparation should use the cell class for which trajectory and relay measurements are strongest: neutrophils in a transparent or microfluidic wound model. This establishes whether the analysis can separate source gradients, self-generated relay, and neural or neuropeptide modulation. A T-cell experiment follows only after a tissue and T-cell subtype with a documented migratory and neural interaction route is selected.

This sequence preserves the original white-blood-cell problem while preventing a neutrophil result from being marketed as a T-cell result.

8.2 Joint measurement

For each identified mobile cell, record:

- three-dimensional position, polarity, speed, turning angle, arrest, and contact history;
- wound-field reporters or calibrated proxies;
- relay signals such as LTB4 and cellular calcium activity where appropriate;
- local neural activity and neuropeptide concentration, including CGRP when the preparation supports it;
- receptor expression or occupancy proxies;
- cell calcium, membrane, metabolic, and morphology state;
- tissue geometry, flow, oxygen, temperature, and mechanical constraints.

Clock synchronization is essential. A phase relation estimated from independently drifting acquisition systems is an instrumentation artifact, not a biological result.

8.3 Perturbations and ablations

The nested models correspond to experimental ablations:

- remove or flatten the wound field;
- block immune relay while preserving the wound source;
- block the selected neural or neuropeptide route while preserving measured chemical fields;

- perturb the receiver receptor or intracellular state predicted to compute the residual;
- restore the blocked carrier or receiver function at the predicted point.

The decisive test is residual course correction. If measured gradients and relay waves already explain turning, adding phase or spectral variables should not be rewarded for overfitting. Conversely, if M_3 predicts held-out turns that M_2 misses, the same advantage must disappear when the proposed phase-bearing carrier or receiver mechanism is interrupted.

8.4 Negative result that would improve the theory

A high-quality negative result is informative. Suppose neural perturbation changes total recruitment or inflammatory phase but not the direction of moment-to-moment turns after chemical fields are controlled. The conclusion would be that the neural route modulates immune gain or phenotype rather than navigation. That result would refine the body–brain network without preserving an unsupported steering mechanism.

9 Application proof: a route-recovery and navigation benchmark

Before collecting expensive multimodal biological data, build an executable simulator with known ground truth. It should generate:

- diffusing and decaying wound fields;
- self-limiting mobile-cell relay waves;
- neural or neuropeptide gain modulation;
- receptor adaptation and state-dependent sampling;
- realistic localization error, dropped frames, clock drift, and measurement noise;
- agents governed by M_0, M_1, M_2, or M_3.

The benchmark asks whether the analysis can recover which mechanism generated each dataset. It must include adversarial cases in which a common unmeasured field creates spurious phase alignment, neural activity follows rather than leads migration, or a slow reporter makes one channel appear predictive.

For true model label Z and recovered label $\widehat{Z}$, report confusion matrices, calibration, parameter recovery, trajectory error, and route-edge false discovery. A useful identifiability score is

$$I_{\text{route}} = 1 - \frac{\mathcal{E}_{\text{recovered}}}{\mathcal{E}_{\text{null}}}, \tag{14}$$

where $\mathcal{E}$ combines route-classification and parameter-recovery error under a prespecified weighting. The exact metric will be frozen before simulation. Success on synthetic data shows that the proposed analysis can recover a mechanism under declared assumptions. It is not evidence that biology uses that mechanism.

10 Predictions and falsifiers

The typed network makes distinct predictions.

1. **Receiver-state prediction.** The same cytokine or wound signal will produce different downstream consequences as receptor state, baseline activity, or inflammatory history changes. Huerta and colleagues already demonstrate this form of context dependence in vagal sensory responses [12].
2. **Route-delay prediction.** Chemical diffusion, neural conduction, cell migration, and transcriptional response will occupy separable delay distributions. A model that forces them into one timescale will misorder causes.
3. **Relay prediction.** Blocking an immune relay will change spatial recruitment propagation differently from flattening the original wound field.
4. **Neural-modulation prediction.** Blocking a local neuropeptide route will alter recruitment, survival, or phenotype according to tissue phase even when the primary wound field remains.
5. **Receiver-relative prediction.** If the SAN extension is real, residual timing or spectral variables will improve held-out prediction of turns beyond M_2, and the gain will be receiver-state specific.
6. **Selective-loss prediction.** Interrupting the proposed carrier or receiver mechanism will remove the M_3 advantage while leaving at least part of gradient-based navigation intact.

The strongest falsifier is straightforward: across sufficiently powered, synchronized experiments, M_3 adds no reproducible held-out information beyond measured gradients, relays, neural variables, and receiver state. That result rejects the receiver-relative phase or spectral navigation term for the tested preparation while leaving the typed body–brain architecture and established neuroimmune routes intact.

11 Significance and limits

The scientific gain is not a claim that one formula unifies immunology and neuroscience. It is a disciplined common interface. Every cell and circuit can be described by what it receives, how its current state transforms that input, what it emits, which receiver changes next, and what delay and uncertainty intervene. That abstraction makes cross-scale hypotheses comparable without erasing their physical differences.

This approach also changes how medical consequences are interpreted. Immune support of memory, immune influence on anxiety, brain retrieval of inflammatory state, neural regulation of cytokines, and immune-cell navigation are not interchangeable evidence. Their value emerges when they are connected as a causal graph and each edge is independently measured.

The receiver-relative extension is the riskiest and therefore most discriminating part of the program. Cellular oscillations are common, but a measurable oscillation is not automatically an address. Phase estimates can be contaminated by reporter kinetics, sampling rate, tissue motion, shared drive, and post hoc frequency selection. The experiment must first demonstrate a stable oscillatory variable, then show directional information beyond nonoscillatory features, then selectively remove that information through intervention.

The first studies will necessarily be animal, ex vivo, or synthetic. Human relevance requires independent replication, appropriate tissue evidence, and noninvasive or clinically justified measurements. No therapeutic recommendation follows from the present model.

12 Parameter and path-certificate specification

Every fitted or simulated quantity is bound to a carrier, cell class, preparation, tissue, timescale, unit, and source locator. A wound-lipid result cannot supply a cytokine coefficient; a neutrophil relay cannot supply a T-cell motility prior; and a vagal calcium trace cannot be silently treated as a phase-coded immune address. The admissible uncertainty set is

$$\Theta_P = \left\{ \vartheta : \underline{\vartheta}_{kP} \le \vartheta_k \le \overline{\vartheta}_{kP},\ k \in K_P \right\}, \tag{15}$$

where (P) denotes the named route and preparation. If the literature supplies no defensible range, the value is `UNKNOWN` and contributes no positive evidence to the route. A sampled or nominal run must not be described as a worst-case route test.

For a complete-path claim, every edge must pass its own intervention and delay check. Let Z_e be the downstream mediator or endpoint for edge (e). A path certificate is emitted only when

$$\text{Cert}(P) = 1 \iff \bigwedge_{e \in P} \left[\Delta Z_e^{do(e)} \neq 0 \ \wedge\ \tau_e \in \mathcal{T}_e \right]. \tag{16}$$

With cell-class, tissue, direction, and endpoint-specific controls, a complete path may remain unresolved even when several individual edges are established; the missing edge is reported rather than filled by analogy. The receiver-relative extension must also beat a route-matched baseline on held-out data and lose its advantage when the named upstream carrier or receiver state is selectively perturbed.

12.1 The experimental unit comes before the estimator

Live imaging can produce millions of cell-time observations from a handful of animals, dishes, wounds, or imaging sessions. Those observations increase temporal and spatial resolution; they do not automatically increase the number of independent interventions. Lazic, Clarke-Williams, and Munafò distinguish biological units, experimental units, and observational units and show why treating subsamples as independent replicates inflates evidence [23]. The present design therefore indexes each observation explicitly:

$$O_{pait} = \left(\mathbf{c}_{pait}, \mathbf{r}_{pait}, \mathbf{n}_{pait}, \mathbf{z}_{pait}, \mathbf{X}_{pait}, Y_{pait} \right), \tag{17}$$

where p is a preparation or experimental run, a is the independently assigned animal, tissue unit, chamber, or culture, i is a tracked cell, and t is time. The identity of the experimental unit is determined by the assignment mechanism. If treatment is assigned to an animal, cells and frames are observational subsamples. If a microfluidic chamber is assigned as one unit, cells sharing that chamber are not independent treatment replicates. Repeating one chamber or one animal over time cannot establish population generality.

This hierarchy changes every inferential claim. Cell-level models may learn trajectories, but uncertainty about an intervention is computed over independently assigned experimental units. The primary analysis reports both the number of experimental units and the much larger number of observational units. Attrition, excluded tracks, and failed reporter channels are reported at both levels.

12.2 Timescale-separated route kernels

The route classes do not operate on one clock. Neural conduction and synaptic transmission can change within milliseconds to seconds; receptor binding, calcium responses, polarization, and turning can evolve over seconds to minutes; relay fields and local recruitment can evolve over minutes; transcriptional, phenotypic, systemic, trafficking, memory, and clinical endpoints can extend from hours to days or longer. These are order-of-magnitude planning bands, not transferable constants.

For edge e, let $h_e(\tau)$ be a preparation-specific response kernel supported on the preregistered delay window $\mathcal{T}_e$. The receiver exposure is

$$q_{e,pait} = \int_{\tau \in \mathcal{T}_e} h_e(\tau)\, x_{e,pai}(t-\tau)\, d\tau. \tag{18}$$

An edge is temporally inadmissible when the measured effect precedes its candidate cause outside the clock-error envelope, or when the fitted kernel places material mass outside $\mathcal{T}_e$. Reporter impulse responses are convolved into the observation model rather than treated as biology. Clock offsets are estimated from common synchronization pulses and then frozen; a freely optimized lag is not evidence for a route.

The delay ledger uses five noninterchangeable bands:

Band	Planning scale	Examples	Primary confound
T0	milliseconds–seconds	action potentials, synaptic relays, fast vagal responses	channel latency and clock offset
T1	seconds–minutes	receptor activation, calcium response, polarity, turning	indicator kinetics and tissue motion
T2	minutes–hours	relay propagation, recruitment, mediator accumulation	diffusion, perfusion, and shared wound severity
T3	hours–days	transcription, phenotype, retention, behavioral adaptation	stress, sickness, circadian phase, and batch
T4	days–months	tissue persistence, memory, disease or treatment endpoints	attrition, co-treatment, natural history, and selection

No coefficient is transported from one band to another without an explicit measurement bridge.

12.3 Predictive hierarchy and the causal boundary

Let G, R, N, and Z denote measured wound-gradient, immune-relay, neural/neuropeptide, and receiver-state histories. Let U denote remaining common causes such as wound severity, perfusion, temperature, tissue geometry, stress, and batch. A transparent structural template is

$$Y_{pait} = g(G_{pait}, R_{pait}, N_{pait}, Z_{pait}, U_{pa}, t) + b_{pa} + \varepsilon_{pait}, \tag{19}$$

where b_{pa} is an experimental-unit effect. Observational prediction cannot by itself distinguish $N \to Y$ from $U \to \{N, Y\}$, delayed reporter leakage, or $Y \to N$. The manuscript therefore uses two levels of language:

- **predictive contribution:** a feature improves held-out scoring in experimental units never used to fit or tune the model;

- **causal route contribution:** a randomized or otherwise justified intervention changes the named mediator and endpoint in the predicted temporal order, while route-matched controls and downstream rescue constrain alternatives.

For a route effect ψ_e, point identification requires a declared assumption set:

$$\begin{aligned}\mathcal{A}_e = \{&\text{consistent intervention}, \quad \text{positivity}, \quad \text{assignment exchangeability},\\ &\text{declared interference}, \quad \text{measured timing}, \quad \text{tool-footprint bounds}\}.\end{aligned} \tag{20}$$

If any required assumption is untested or implausible, the route is labeled partially identified or predictive-only. In particular, immune-cell relays violate naive no-interference assumptions by design. The unit of interference must therefore be specified—for example, a wound, chamber, animal, or local neighborhood—and the estimand changed accordingly. A matched rescue narrows explanations but does not prove exclusivity when the rescued mediator has parallel targets.

12.4 Receiver-relative innovation, coefficient of variation, and phase are separate variables

The strongest SAN proposal is first stated without specialized vocabulary. A cell has a current receptor and intracellular state. It receives a new combination of chemical, cellular, and neural inputs. The scientifically relevant difference may be the part of that new combination not predicted from the cell's measured recent history and state. That difference is then tested for its effect on the cell's next course correction.

Define the cross-fitted innovation

$$\boldsymbol{\epsilon}_{pait} = \mathbf{f}_{pait} - \widehat{\mathbb{E}}^{(-pa)}[\mathbf{f}_{pait} \mid \mathcal{H}_{pai,t-1}, Z_{pait}, G_{pait}, R_{pait}, N_{pait}] \,, \tag{21}$$

where the expectation is trained without experimental unit (p, a). This prevents a cell's own future or near-duplicate frames from teaching the model the residual later used to score that cell.

Three quantities remain distinct:

$$CV(x) = \frac{s(x)}{|\bar{x}| + \epsilon}, \qquad R_\phi = \left| \frac{1}{K} \sum_{k=1}^{K} e^{i\phi_k} \right|, \qquad \rho = 1 - \cos(\phi_W - \phi_R - \theta_Z). \tag{22}$$

The coefficient of variation describes relative dispersion of a suitably scaled quantity. R_ϕ describes circular concentration. ρ is one receiver-relative phase mismatch. None is Shannon information by definition, and none is a phase-wave differential merely because it varies. A phase term enters the primary model only when a preregistered signal has reproducible spectral concentration, phase reliability above the calibration floor, adequate sampling, and a biologically plausible delay. Otherwise the preregistered analysis tests the nonoscillatory innovation in equation (21) and records the phase branch as unavailable.

Only after this operation and its measurement conditions are clear is the SAN name earned: a **phase-wave differential** is the receiver-relative, causally consequential departure when the measured departure is specifically phase-structured. COT is the broader receive–transform–transmit account; PWD is a candidate discriminating variable inside it, not a substitute for chemotaxis or neuroimmunology.

12.5 Nested cross-fitting and experimental-unit uncertainty

Model selection, tuning, phase-band choice, lag choice, and normalization are confined to development and calibration units. Evaluation units remain sealed. For model M_k, parameters in outer fold a are

$$\widehat{\theta}_k^{(-a)} = \arg\min_{\theta} \sum_{a' \neq a} \sum_{i,t} \ell\big(Y_{pa'it}, f_k(O_{pa'it}; \theta)\big) + \lambda_k \|\theta\|_2^2, \tag{23}$$

with λ_k selected inside the training units. The incremental score for held-out experimental unit a is averaged inside that unit before population inference:

$$\Delta_a^{(k,k-1)} = \frac{1}{n_a} \sum_{i,t \in a} \left[\ell_{k-1}(Y_{ait}) - \ell_k(Y_{ait})\right]. \tag{24}$$

The primary uncertainty object is the vector of A independent-unit scores, not the vector of cell-time residuals. A cluster-level standard error is

$$\widehat{SE}_{\text{EU}}(\bar{\Delta}) = \sqrt{\frac{1}{A(A-1)} \sum_{a=1}^{A} \left(\Delta_a - \bar{\Delta}\right)^2}. \tag{25}$$

Confidence intervals use experimental-unit resampling or randomization consistent with assignment. When the number of independent units is too small for a stable asymptotic interval, the paper reports the full unit-level distribution and exact assignment-compatible randomization result rather than a cell-level pseudo-precision. Sex, age, batch, litter, cage, preparation day, tissue, and operator are blocked or modeled according to how assignment occurred; none is automatically an independent replicate.

The primary model-comparison family contains the three ordered increments $M_1 - M_0$, $M_2 - M_1$, and $M_3 - M_2$ for each preregistered endpoint. Multiplicity is controlled over that frozen family. Secondary frequency, lag, cell-subtype, and tissue analyses are clearly labeled exploratory and cannot rescue a failed primary test.

12.6 Identifiability tests and negative-control fixtures

The identifiability design requires each favorable interpretation to survive a corresponding failure fixture.

Candidate explanation	Required discriminator	Adverse or null outcome retained
wound field drives turning	source flattening or source displacement	no change, reverse effect, or effect only on speed
immune relay adds direction	relay block with wound field preserved	recruitment magnitude changes but direction does not
neural route leads immune change	neural interruption plus clock-verified lag	neural activity follows motion or shares a common cause
receiver state transforms input	receptor/state perturbation and state-stratified prediction	pooled gain disappears within state strata
phase-relative term adds information	fixed-band held-out gain and carrier-specific loss	gain vanishes after reporter deconvolution or cyclic shift
complete path is mediated	sequential edge perturbation and matched downstream rescue	incomplete, nonspecific, paradoxical, or over-rescue

Four mandatory negative controls are analyzed with the same pipeline:

1. **Time reversal or illegal lead:** future movement is not permitted to predict the candidate upstream carrier in the causal model.
2. **Within-unit cyclic shift:** aligned timing is destroyed while marginal spectra and autocorrelation are preserved as far as the shift permits.
3. **Reporter-kernel control:** a slow indicator is deconvolved or simulated so its lag cannot create a false lead.
4. **Shared-driver control:** wound severity, perfusion, temperature, motion, and treatment batch are included or experimentally balanced so one latent field cannot impersonate two route edges.

For each route e, define prediction, timing, intervention, and rescue indicators $P_e, T_e, I_e, R_e \in \{0, 1, \mathrm{NA}\}$. The honest certificate is

$$\mathrm{Cert}_e = \begin{cases} \text{complete}, & P_e = T_e = I_e = R_e = 1, \\ \text{partial}, & \text{some but not all required tests pass}, \\ \text{refuted-in-preparation}, & \text{a preregistered directional test fails}, \\ \text{unresolved}, & \text{a required test is missing or nonidentifiable.} \end{cases} \tag{26}$$

`NA` never becomes a pass. A complete organism-level path requires every required edge to be complete for the named preparation and endpoint. A failure of the phase-relative edge does not erase established chemical, relay, vagal, neuropeptide, or immune-to-neural routes.

12.7 Intervention matrix, power, and stopping rules

The smallest decisive experiment is a factorial route test, not an omnibus intervention. For wound navigation, the core factors are wound source present/flattened, relay intact/blocked, neural or neuropeptide route intact/blocked, and receiver mechanism intact/perturbed. Sham procedures, vehicle, reporter-only, and tool-footprint controls are crossed where feasible. The analysis asks whether the increment associated with each route disappears under its matched interruption while unrelated retained routes remain measurable.

For the body–brain path, the ordered perturbation is cytokine/receptor → afferent population → brainstem population → descending effector → immune endpoint. Rescue occurs immediately downstream of the interrupted edge. A behavioral or disease endpoint is secondary to the mediator chain; clinical improvement alone cannot certify every edge.

Power is based on experimental-unit variation in the held-out score or intervention contrast. If δ_* is the smallest scientifically relevant unit-level increment and s_Δ is estimated from a blinded pilot, an initial planning approximation is

$$A \geq \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{\delta_*/s_\Delta} \right)^2, \tag{27}$$

followed by design-specific simulation that includes clustering, missing tracks, intervention failure, and multiplicity. The protocol does not invent a universal animal number before those quantities exist. The stopping rules are frozen before unblinding: technical stop for failed clock or reporter calibration; safety stop for prespecified welfare criteria; futility stop only through a declared group-sequential rule; and scientific stop when the required route variable is physically unmeasurable in the selected preparation.

12.8 Medical, safety, and translation boundary

This paper is a mechanistic and methods proposal. It does not recommend vagus stimulation, cytokine manipulation, immune-cell transfer, receptor blockade, wound induction, or any intervention outside approved research and clinical governance. The human RESET-RA result [26] shows why this boundary matters: a regulated device trial can provide clinical evidence while leaving route-level mechanistic questions open, and implantation-related risks remain part of the evidence.

Animal work requires ethical review, reduction through simulation and efficient factorial design, refinement of imaging and perturbation procedures, and replacement whenever a culture or synthetic test can answer the same question. Human tissue and in-vitro models require donor, consent, biosafety, and generalization boundaries. Sex, age, immune history, medication, pathogen status, and disease state are biological moderators, not nuisance variables to be erased.

The formal-identifiability specification reaches a design gate, not biological validation. Its next honest gate is an executable synthetic reference application that instantiates the frozen models, confounds, interventions, clustered units, and adverse outcomes. Synthetic recovery can validate code and reveal nonidentifiability under declared assumptions; it cannot show that living cells use the proposed receiver-relative mechanism.

13 A staged biological protocol that can decide the route

13.1 Why the experiment must be staged

The strongest form of the proposal is not served by one enormous experiment that combines a wound, every immune-cell class, neural recording, cytokine profiling, behavior, and a phase analysis. Such an omnibus design would make a failure uninterpretable. A missing effect could reflect the wrong cell class, a failed reporter, an unsuitable tissue, inadequate synchronization, an ineffective perturbation, or a genuinely absent route. The biological program therefore separates capability gates from mechanism gates.

The navigation program proceeds through five ordered stages:

Yang, Collins, and Meyer showed in neutrophil-like cells that local Cdc42 activity can precede turning, while Beerman and colleagues demonstrated leading-edge calcium flux and reversible calcium manipulation in zebrafish neutrophils [27,28]. Khazen and colleagues linked neutrophil calcium activity to LTB4-dependent recruitment in a mouse clustering preparation [32]. These results make N1 and N2 experimentally grounded without treating Cdc42 or calcium as the SAN result. The new question begins only after those established steering and relay variables are measured.

T cells require a separate stage. Dong and colleagues measured Orai1-dependent calcium transients accompanying pauses and altered motility in human and mouse T-cell preparations [29]. That result justifies a T-cell receiver-state measurement; it does not establish wound steering, a neural address, or a phase-wave differential.

Stage	Preparation	Question decided	Promotion requirement
N0	cell-free gradient and clock phantom	Can the instruments recover known spatial fields, delays, and phase relations?	calibrated gradient, clock-error, localization, and reporter-kernel bounds
N1	neutrophil-like PLB-985 cells in a microfluidic or photorelease preparation	Do local receptor, Cdc42, or calcium states precede and predict turning beyond the imposed gradient?	reproducible turning, usable tracks, and held-out M0 scoring
N2	primary neutrophil relay preparation	Does an immune relay add direction or recruitment dynamics beyond the primary field?	selective relay interruption with the wound or source field retained
N3	route-matched sensory-neuron or neuropeptide preparation	Does the neural variable alter direction, gain, survival, or phenotype, and which of those endpoints changes?	carrier measurement, receptor evidence, matched interruption, and retained non-neural fields
N4	selected T-cell subtype and tissue route	Does the same analysis generalize to a named T-cell system?	prior evidence for that cell, receptor, tissue, and neural or neuropeptide route

13.2 Frozen navigation endpoint and factorial route test

The independently assigned unit in N1 and N2 is the microfluidic chamber, photorelease field, fish, tissue preparation, or other smallest unit that independently receives a route intervention. Cells and frames nested within it are observational units. The primary endpoint is the experimental-unit mean held-out log-score increment for the newest route under test. For the receiver-relative stage,

$$\Delta_a^{(3,2)} = \frac{1}{n_a} \sum_{i,t \in a} \left[\log p(\Delta \mathbf{X}_{ait} \mid M_3) - \log p(\Delta \mathbf{X}_{ait} \mid M_2) \right]. \tag{28}$$

Angular error, speed, arrest, recruitment count, survival, polarity, morphology, and phenotype are secondary or route-specific endpoints. A decrease in angular error with no change in total recruitment is a steering result; a change in recruitment with no angular effect is a gain result. Neither is substituted for the other.

The navigation intervention vector is

$$\mathbf{A}_a = (A_G, A_R, A_N, A_Z), \tag{29}$$

where A_G alters the primary gradient, A_R alters the immune relay, A_N alters the neural or neuropeptide route, and A_Z alters the proposed receiver mechanism. Each factor has a matched sham and a measured tool-footprint control. The smallest decisive design does not require every 2^4 combination at once. It requires the intact condition, each single matched interruption, the prespecified interactions needed for route ordering, and downstream rescue where technically justified. Combination arms that lack positivity, welfare justification, or a distinct scientific contrast remain absent and cannot be imputed as passes.

Randomization occurs at the experimental-unit level and is blocked by preparation day, instrument, operator, sex where animals are used, and other variables known before assignment. Allocation is concealed from track curation and primary analysis. Inclusion and exclusion rules are frozen before outcome access. Failed chambers, lost synchronization pulses, reporter saturation, dead cells, truncated tracks, and intervention failures remain in an attrition ledger with their treatment assignment masked during adjudication.

13.3 Clock, reporter, and phase measurement contract

The observed reporter is not the latent biological signal. For channel (j),

$$y_j^{obs}(t) = \int k_j(\tau) y_j^{bio}(t-\tau)d\tau + \nu_j(t), \tag{30}$$

where k_j is the measured indicator and acquisition kernel. Genetically encoded calcium indicators can have materially different rise, decay, and isomerization kinetics [30]. The kernel is therefore estimated from calibration data or bounded from an external calibration performed in the same temperature, expression, and acquisition regime. A favorable lag that disappears after the reporter model is applied is recorded as an imaging explanation, not a biological lead.

Every acquisition device receives a shared synchronization event before and after each run. If $\widehat{t_j}$ is a channel timestamp and (t) the reference clock, the maximum residual offset is

$$\delta_{clock} = \max_{j,t} |\widehat{t}_j(t) - t|. \tag{31}$$

A candidate causal delay is inadmissible when its lower bound does not exceed the combined clock, exposure, registration, and reporter uncertainty envelope. Drift correction is learned from calibration pulses and then frozen. Biological events cannot be used to retune the clocks that will later be tested for temporal order.

The phase branch is available only when all preregistered conditions pass:

$$G_{phase} = \mathbb{1}\left[Q_{spec} \geq q_*,\ Q_{phase} \geq r_*,\ f_s \geq 2\kappa f_{max},\ \delta_{clock} < \delta_*,\ k_j \text{ identified}\right]. \tag{32}$$

Here Q_{spec} is spectral concentration, Q_{phase} is repeat or split-half phase reliability, f_s is effective sampling frequency, and $\kappa > 1$ is a prespecified safety factor above the Nyquist minimum. Thresholds are fixed from N0 and development units. If $G_{phase} = 0$, the nonoscillatory cross-fitted innovation can still be tested, but the PWD label is unavailable for that preparation.

13.4 Track construction and missingness

Automated segmentation and tracking are tuned only on development movies. A track is eligible when it satisfies frozen minimum duration, localization confidence, field-of-view margin, and viability rules. Track linking may not use treatment identity or the future outcome under evaluation. Manual corrections are logged and adjudicated blind.

Missingness is classified before analysis:

- **technical:** focus loss, detector saturation, clock failure, segmentation or registration failure;
- **geometric:** exit from the field, occlusion, or collision;
- **biological:** arrest, adhesion, death, tissue entry, or phenotype change;
- **administrative:** protocol deviation or unavailable assay.

Because biological dropout can itself be an endpoint, complete-case analysis is never the sole result. The primary analysis reports track retention by arm and unit, a worst/best bounded sensitivity analysis for truncated directional endpoints, and a joint secondary model of motion and dropout when its assumptions are defensible. If attrition destroys overlap among arms, the affected route certificate is unresolved.

13.5 Body-brain route protocol

The body-brain experiment is also staged. B0 verifies cytokine assays, neural reporters, stimulation timing, and the downstream immune endpoint separately. B1 identifies the afferent cytokine-sensitive population and its brainstem receiver. B2 tests the descending effector and peripheral immune endpoint. B3 performs matched downstream rescue. Behavioral and clinical-style endpoints enter only after the mediator chain is measurable.

For ordered edge (e), let M_e be its downstream mediator and Y the route endpoint. The preregistered edge contrast is

$$\psi_e = \mathbb{E}[M_e, Y \mid do(A_e = 1)] - \mathbb{E}[M_e, Y \mid do(A_e = 0)], \tag{33}$$

evaluated as separate mediator and endpoint contrasts with their own delay windows. A behavioral change without the mediator change is a behavioral finding and a route-certificate failure. A mediator change without the endpoint is a partial edge. Rescue must restore the prespecified downstream mediator and endpoint without restoring unrelated tool-footprint controls.

Generic sickness, pain, locomotion, temperature, feeding, corticosterone, circadian phase, and handling remain measured alternatives. They are not covariates chosen after seeing which one protects a favorable route. The adjustment set is defined from the causal graph before unblinding; variables downstream of the intervention are not casually conditioned away.

13.6 Power is an assurance calculation over experimental units

The protocol does not set a universal animal or chamber count. A blinded pilot estimates the experimental-unit distribution, intervention success, track yield, missingness, and measurement reliability. Candidate designs are then simulated under the smallest scientifically relevant effect δ_*, the observed cluster structure, the frozen multiplicity family, and the planned analysis.

For design (d), define assurance

$$\mathcal{A}(d) = \Pr_{\theta \sim \Pi_{pilot}} [\text{primary decision is correct and calibration passes} \mid d, \theta] \,. \tag{34}$$

The selected design is the smallest ethically acceptable (d) whose lower uncertainty bound for $\mathcal{A}(d)$ exceeds the preregistered target. The calculation includes whole-unit loss, failed perturbations, sex and batch blocks, multiplicity, and the possibility that the phase branch is unavailable. Cell-time rows never replace experimental units in this calculation.

ARRIVE 2.0 requires explicit reporting of study design, experimental unit, sample-size rationale, inclusion and exclusion, randomization, blinding, outcomes, statistical methods, animal characteristics, procedures, and results [31]. Those items are treated here as design-time commitments rather than prose added after the experiment.

13.7 Route decisions and adverse outcomes

The primary decisions are directional and bounded:

Observation	Navigation interpretation	Certificate state
M_1 improves over M_0, relay block removes the gain	relay contributes in the named preparation	partial or complete relay edge, depending on rescue and timing
neural interruption changes recruitment but not held-out turns	neural gain or phenotype effect, navigation null	neural modulation edge partial; steering refuted in preparation
M_3 gain vanishes after reporter correction	reporter kinetics explain the apparent lead	receiver-relative phase edge refuted in preparation
M_3 gain survives held-out scoring but not matched receiver perturbation	predictive association without receiver mediation	partial or unresolved
phase reliability fails	phase is not measurable under the frozen setup	phase-branch-unavailable
matched interruption removes M_3 gain and downstream rescue restores it selectively	receiver-relative route supported in the named preparation	eligible for complete edge review
undercoverage, wrong sign, reverse timing, or failed rescue	adverse result	retained without reinterpretation as support

No route result is transferred between neutrophils, T cells, macrophages, tissues, species, disease states, or timescale bands without a new measurement bridge.

13.8 Ethics and translational stop rules

N0 and the executable synthetic application precede living preparations. N1 favors cell models and controlled gradients before animal escalation. Primary-cell and animal stages proceed only when the previous stage establishes instrument capability and the next stage answers a question that the prior stage cannot. Welfare, biosafety, donor, and clinical-device governance remain independent of statistical success.

Technical stopping occurs for failed timing, gradient, registration, reporter, or intervention calibration. Biological stopping occurs under approved welfare criteria. Statistical futility uses only the frozen group-sequential rule. A scientifically inconvenient null is not a technical failure and does not trigger replacement of the endpoint.

13.9 Prospective biological-protocol boundary

The prospective protocol freezes an experiment that can separate gradient sensing, cellular relay, neural modulation, receiver-state transformation, and a genuinely phase-structured residual. It adds no biological result. The next gate is the executable synthetic reference application specified in Section 9, using the experimental unit, clocks, reporter kernels, negative controls, intervention states, and adverse outcomes fixed here.

14 Executable route recovery under known synthetic mechanisms

14.1 What the application was built to decide

The frozen synthetic application asks a narrower question than the proposed biology: if data are generated by the four route classes defined in equation (11), can the analysis recover the generating class on experimental units it did not fit, recognize selective interruption and rescue, and reject familiar false-positive mechanisms? This is a code and identifiability test. The simulated chamber is the independent unit; its eight cells and thirty-two time points per cell are observational subsamples.

The four generators are deliberately nested but distinct:

- M_0 generates motion from a wound-gradient vector and receiver state;
- M_1 adds an immune-relay vector;
- M_2 adds a neural or neuropeptide vector whose effect depends partly on receiver state;
- M_3 adds a receiver-relative innovation and a phase vector.

Each candidate model is fitted only on sixteen development units generated by its declared mechanism. Eight independent calibration units per route select a single temperature for class probabilities. Sixteen sealed evaluation units per route supply the final route test. There are 64 development, 32 calibration, and 64 evaluation units, each containing 256 cell-time observations.

For candidate route k, the unit receives a mean Gaussian predictive score s_{ak}. The calibrated route probability is

$$\widehat{p}_{ak} = \frac{\exp(s_{ak}/T)}{\sum_{j=0}^{3} \exp(s_{aj}/T)}, \tag{35}$$

where T is selected only on calibration units. No evaluation label changes a coefficient, feature, seed, threshold, or temperature.

14.2 Route-classification result

The matched synthetic evaluation classified all 64 units correctly. Accuracy, macro-F1, and each route's recall were 1.000. The ten-bin expected calibration error was (1.10\times10^{-6}), with temperature (T=0.05). The confusion matrix was diagonal with sixteen units in each class.

This is a capability result under strongly separated, matched generators. It is not an estimate of how well the method will classify living biology, a new laboratory, an unseen generator family, or a preparation whose true mechanism is a mixture of the four classes. Perfect synthetic separation is therefore not presented as biological probability or clinical confidence. It mainly proves that the implemented features, fitting, scoring, calibration, and sealed-unit routing are internally coherent in the domain they were designed to recognize.

14.3 Matched interruption and rescue expose a nonselectivity failure

For routes M_1 through M_3, the application regenerated each evaluation unit under intact, matched-block, 75-percent rescue, and unrelated-gradient-block conditions using the same unit seed. The route advantage over its immediate predecessor is

$$A_{ak} = s_{ak} - s_{a,k-1}, \qquad D_k = \text{median}(A_{ak}^{intact}) - \text{median}(A_{ak}^{block}). \tag{36}$$

The rescue fraction is

$$F_k = \frac{\text{median}(A_{ak}^{rescue}) - \text{median}(A_{ak}^{block})}{D_k}. \tag{37}$$

Route	Intact median	Matched block	Rescue	Unrelated gradient block	Selective drop D_k	Rescue fraction F_k	Synthetic certificate
M1 over M0	1.786	-1.852	0.881	-0.124	3.638	0.751	partial or refuted
M2 over M1	0.904	-0.910	0.470	-0.528	1.814	0.761	partial or refuted
M3 over M2	5.148	-5.798	2.325	5.135	10.947	0.742	complete in the synthetic generator only

All three matched interruptions removed the newest-route advantage and all three rescues restored more than the frozen 0.50 fraction. However, the unrelated gradient block also erased the M1 and M2 advantages. Those two route certificates therefore fail the selectivity requirement despite their excellent intact classification. This adverse result is not a nuisance. It shows that the fitted relay and neural templates depend on the baseline gradient regime in this generator. The generator-shift audit must test domain shift and interaction structure rather than treating route increments as portable additive modules.

M3 retained its advantage after the unrelated gradient block. Its certificate is still limited to this synthetic generator. A real receiver-relative edge would require the biological timing, measurement, intervention, and rescue evidence frozen in the prospective biological protocol.

14.4 Adversarial fixtures

Five separately generated fixtures tested whether apparently favorable prediction would be overinterpreted:

Fixture	Raw apparent gain	Gain or reliability after required control	Correct disposition
shared driver	2.028	0.0169 after shared-driver proxy	nonidentifiable
reverse timing	0.654	0.000097 under legal history	refuted in preparation
reporter lag	0.556	0.000466 after reporter correction	refuted in preparation
unreliable phase	0	0.00713 repeat-phase reliability	phase-branch-unavailable
nonselective tool	0.607 apparent loss	0.345 retained-route selectivity	partial

The controlled-to-raw ratio

$$Q_f = \frac{|G_f^{controlled}|}{|G_f^{raw}| + \epsilon} \tag{38}$$

is diagnostic only when the control corresponds to the known fixture. A small Q_f for the shared-driver or reporter fixture demonstrates that the control can remove the planted artifact; it does not guarantee removal of every latent common cause or every reporter distortion in biological data. The phase fixture is different: its 0.00713 repeat-phase reliability fails the measurement gate before a route effect is scored.

14.5 Reproducibility and the failed first execution

The scientific outputs were regenerated in a separate replay directory and all result files were byte-identical. The run preserved the frozen seed, route definitions, development/calibration/evaluation split, ridge penalty, temperature grid, interventions, fixtures, and thresholds.

The first frozen runner stopped before producing any scientific output because a Python boolean was written with JSON syntax. That failed runner and contract remain preserved. A versioned runner corrected only the execution literal, a second frozen contract bound its hash, and all pre-outcome contract tests were rerun before the successful execution. This engineering failure is included in the companion audit because executable evidence includes failed attempts and repair provenance.

14.6 What the application proves and does not prove

The application establishes four bounded results:

1. the implemented pipeline can distinguish the four strongly separated matched generators on held-out synthetic chambers;
2. matched interruption and partial rescue produce the expected score changes;
3. the pipeline refuses complete certificates when an unrelated intervention destroys the advantage;
4. the declared negative controls detect the five planted artifact families.

It does not establish that immune cells use the M_3 mechanism, that neural activity steers T cells, that a phase-bearing variable exists in the selected preparation, or that the synthetic noise and interaction structure approximate living tissue. The final biological test remains sealed and unopened.

14.7 Synthetic reference-application result

The synthetic reference application passes its bounded executable gate. The next honest gate is not a larger favorable simulation. The generator-shift audit therefore stresses the result with unseen generator families, mixtures of route mechanisms, coefficient and sign variation, correlated fields, altered gradient geometry, stronger dropout, weaker phase reliability, undercoverage tests, simpler and more flexible baselines, and ablations of each feature family. M1 and M2 selectivity failures are the first target of that expansion.

15 What survives when the synthetic world changes

15.1 The problem before the robustness vocabulary

A route-recognition system can look decisive when the world that tests it is built by the same rules as the world that trained it. Living immune navigation is less accommodating. Signal strengths vary; a field can reverse its behavioral effect; two candidate fields can rise together; tissue geometry can bend the relation between a measured gradient and motion; tracks disappear selectively; an oscillatory measurement can become unreliable; and more than one route can operate in the same preparation. The question is therefore not whether the fitted reference application can repeat its original success. The question is which parts of that success remain when those assumptions are changed without changing the fitted model.

The robustness contract froze that question before opening any new outcome. The exact reference-application regression coefficients and temperature were loaded from their preserved files and were not refitted. Ten non-mixture generator families and three mixture families were named in advance. Each non-mixture family supplied twenty-four new chambers per route, for 960 independently generated stress chambers. Each chamber still contained eight cells and thirty-two time steps before any declared dropout. Three classifiers scored the same stress units: the preserved nested Gaussian score classifier, a pooled-covariance classifier using the four frozen model scores, and a sixteen-feature ridge labeler. The latter two learned only from the original reference-application development units and selected temperature only on the old calibration units.

Let a_{sm} be accuracy for scenario s and method m, and let a_{0m} be its new-seed matched accuracy. The descriptive robustness loss is

$$L_{sm} = a_{0m} - a_{sm}. \tag{39}$$

This quantity is not a biological effect size. It shows how much route-label performance changes when a declared generator assumption changes. The frozen primary gate required accuracy of at least 0.60, macro-F1 of at least 0.55, recall of at least 0.40 for every route, and expected calibration error no greater than 0.35. Passing those numerical thresholds could still be refused when the preparation failed an identifiability or measurement gate.

15.2 The frozen model is robust to some shifts and fragile to others

The preserved classifier reproduced perfect classification on the new-seed matched family and on the stronger-coefficient family. It also passed after motion-dependent dropout removed 75 percent of the cell-time rows. That dropout result is useful but bounded: M1 recall fell to 0.50 and M2 recall to 0.625, so it does not justify treating missing tracks as harmless.

The same frozen classifier failed sharply when the newest route contribution was weakened to 0.35 of its original strength or reversed in sign. Both families produced accuracy 0.25, zero recall for M1 through M3, and severe overconfidence. Altering the gradient-to-motion relation to a double-angle field reduced accuracy to 0.479 and eliminated M1 and M2 recall. Nonlinear saturation and the unseen cross-field composition both retained overall accuracy 0.813, but each failed because one route's recall fell below the frozen floor. A favorable average would have concealed those route-specific failures.

The correlated-field family illustrates why prediction and attribution must remain separate. Classification was perfect, yet the maximum measured field correlation was 0.987, above the frozen 0.80 refusal threshold. A system can recognize a planted label while lacking enough independent variation to say which correlated field carried the effect. This family is therefore `nonidentifiable`, not a route certificate.

The weak-phase family makes the complementary point. Its median phase reliability was 0.091, below the frozen 0.30 gate. Classification remained perfect because other receiver-relative and route features remained informative. That success cannot restore the missing phase measurement. The correct disposition is `phase-branch-unavailable`, and no PWD certificate follows from the label accuracy.

Generator family	Accuracy	Macro-F1	Minimum route recall	ECE	Frozen disposition
new-seed matched	1.000	1.000	1.000	<0.000001	pass
newest coefficient 0.35	0.250	0.167	0.000	0.745	adverse
newest coefficient 1.60	1.000	1.000	1.000	0.000001	pass
newest contribution sign-reversed	0.250	0.167	0.000	0.750	adverse
fields correlated at the declared high level	1.000	1.000	1.000	<0.000001	nonidentifiable
double-angle gradient geometry	0.479	0.367	0.000	0.517	adverse
75-percent informative dropout	0.781	0.775	0.500	0.164	pass
weak observed-phase reliability	1.000	1.000	1.000	<0.000001	phase branch unavailable
nonlinear response saturation	0.813	0.782	0.250	0.143	adverse
unseen compositional field	0.813	0.787	0.292	0.147	adverse

15.3 Route increments and multiplicity

For route (k>0), the frozen unit-level increment remains

$$\Delta_{ask} = s_{ask} - s_{as,k-1}, \tag{40}$$

where s_{ask} is the mean predictive score for unit a, stress family s, and candidate route k. A one-sided sign test asks whether more than half of the nonzero increments are positive under the null probability one-half:

$$p_{sk} = 2^{-n} \sum_{j=q}^{n} \binom{n}{j}, \tag{41}$$

with q positive increments among n nonzero units. Thirty route-increment tests and five matched-family feature-ablation tests formed one family. Ordered raw probabilities $p_{(j)}$ were evaluated by the Holm threshold

$$p_{(j)} \leq \frac{0.05}{35 - j + 1}. \tag{42}$$

Twenty-four of the thirty-five tests survived correction; eleven did not. All eleven corrected nulls remain in the adverse ledger. The weak-coefficient and sign-reversal families supplied no favorable route increment. The altered-geometry family retained only the M3-over-M2 increment, severe dropout retained only M3-over-M2, and nonlinear saturation retained M1-over-M0 and M3-over-M2 but not M2-over-M1. The reliable-looking M3 increment in the weak-phase family remains measurement-gated and cannot be reinterpreted as phase evidence.

All five frozen-coefficient ablations reduced matched-family predictive score in all twenty-four tested units and survived Holm correction (adjusted (p=2.09\times10^{-6})). The median score losses were 3.712 for gradient in M0, 1.889 for relay in M1, 0.978 for neural input in M2, 1.766 for innovation in M3, and 1.227 for phase in M3. These are internal matched-generator dependency results. They show that the frozen fitted model used each planted feature family; they do not show that living cells use the corresponding mechanism.

15.4 Stronger baselines change which failures are repairable

Neither comparison baseline dominated every shift. Under altered gradient geometry, all three classifiers failed the frozen gate: accuracy was 0.479 for the preserved classifier, 0.396 for score-LDA, and 0.542 for the sixteen-feature ridge labeler. Under severe dropout, the preserved classifier alone passed all route-specific thresholds. Under nonlinear saturation and the unseen compositional field, however, the ridge labeler achieved accuracy 1.000 and 0.990 respectively, while the preserved classifier failed minimum-recall requirements. These results locate a genuine software-design choice: richer unit-level summaries can recover some nonlinear or compositional shifts, but they do not solve coefficient reversal, geometry change, correlated attribution, or phase measurement.

The ridge result is a development-set comparison, not a post-outcome repair. Its sixteen inputs, penalty, and calibration route were fixed before stress outcomes. The robustness analysis does not replace the primary model with whichever method won a particular family. Such replacement would require a new development-only design, another freeze, and an unopened confirmation partition.

15.5 Mixtures require a different question

A preparation generated jointly by two routes has no honest single-route truth label. The robustness analysis therefore measured whether the most probable label belonged to either planted component, whether the two most probable labels recovered both components, and how dispersed the four probabilities were. Normalized entropy was

$$H_a = -\frac{1}{\log 4}\sum_{k=0}^{3}\widehat{p}_{ak}\log\widehat{p}_{ak}. \tag{43}$$

The equal M1-M2 mixture recovered a component first and both components in its top two for every unit, with median (H=0.441). The equal M0-M3 mixture recovered a component first in only 0.25 of units and both components in the top two in 0.021, with median (H=0.491). The asymmetric M1-M3 mixture chose a component first in every unit but never placed both components in the top two; its median entropy was only 0.007, showing confident compression of a mixture into one route. None of the three families receives a single-origin certificate. Low entropy is not proof that the generator was singular.

15.6 What the robustness gate changes

The combined decision rule can now be stated before its coined terms:

$$C_{sk} = G_{id,s}\, G_{meas,s}\, \mathbb{1}[\text{performance gate passes}]\, \mathbb{1}[\text{selective causal tests pass}], \tag{44}$$

where $G_{id,s}$ denies attribution under excessive field dependence and $G_{meas,s}$ denies phase claims when the phase channel is unreliable. A positive label or score increment cannot bypass either gate. The matched intervention analysis already left M1 and M2 partial or refuted after their unrelated-gradient intervention. The robustness analysis does not erase those adverse certificates; it adds evidence that route recognition is also sensitive to coefficient scale, sign, geometry, nonlinearity, composition, and missingness.

Three of ten non-mixture families passed the complete frozen primary gate. Five met the raw numerical thresholds, but two of those were correctly refused for nonidentifiability or phase unreliability. Twenty-four of thirty-five multiplicity-controlled tests were positive, while eleven remained null. All results, including the three mixture refusals, are preserved in the executable ledger and reproduced byte-for-byte in a second run.

These results strengthen the research program by making its limits operational. They do not validate neural steering of immune cells, a receiver-relative navigation mechanism in living tissue, or a phase-bearing biological signal. They also do not reduce the source-faithful SAN proposal to this first classifier. The proposal asks whether measured receiver history and timing add consequence beyond typed established routes; The robustness analysis shows which synthetic conditions the current implementation can and cannot adjudicate.

15.7 Robustness result and boundary

The generator-shift evaluation passes as a robustness audit because its stress families, baselines, ablations, thresholds, multiplicity method, and refusal rules were frozen before execution; its replay is byte-identical; and its adverse and null results remain prominent. It does not pass as a generally robust route classifier. Weak effects, sign changes, altered gradient geometry, and mixture compression are material failure modes.

The next development stage turned these boundaries into inspectable architecture, causal-route, calibration, confusion, ablation, and trajectory figures. Any estimator revision should then be built only on a new development partition, with explicit mixture recognition, geometry-aware features, uncertainty-aware abstention, and dropout sensitivity. The hash-sealed future final-test partition must remain unopened until that revised architecture, analysis code, and acceptance rules are fixed. The final biological test also remains sealed and unopened.

16 Making the architecture and its failures inspectable

16.1 The problem before the estimator

The generator-shift audit exposed two different errors that a route-recognition system can make. First, it can fail to recognize a route when a coefficient weakens, changes sign, or appears through a different geometry. Second, it can recognize a label while lacking the independent variation or measurement reliability required to attribute that label to a causal route. Those problems are not interchangeable. A better classifier cannot repair a correlated-field attribution problem, and perfect route prediction cannot turn an unreliable observed phase into evidence for a phase-structured mechanism.

The visual architecture therefore begins with physical operations rather than model names. A wound or peripheral state changes local fields. Immune cells relay and terminate signals. A moving receiver samples those fields through its own receptor and intracellular state. Local neural or neuropeptide activity may change recruitment, survival, phenotype, gain, or direction. Vagal and brainstem circuits can receive immune signals and return regulatory output. Movement and repair then change the environment sampled next.

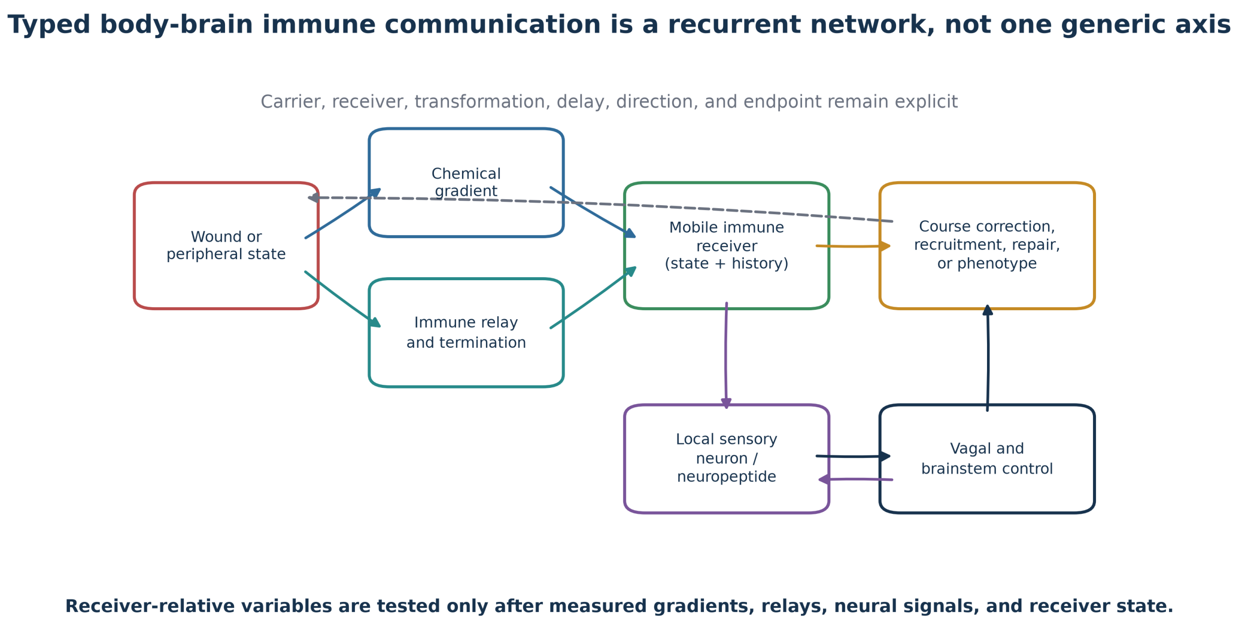

Figure 1: Typed body-brain immune communication architecture. The recurrent architecture separates wound or peripheral sources, chemical gradients, immune relay and termination, the changing mobile immune receiver, local neural or neuropeptide modulation, course correction or phenotype, and vagal or brainstem control. Arrows denote typed candidate routes, not one universal physical channel. The return path emphasizes that action and repair change the next sampled environment.

The architecture is recurrent, but no arrow inherits another arrow's evidence. A zebrafish wound field, a human-neutrophil relay, a mouse CGRP route, a vagal cytokine response, and a T-cell-associated neural endpoint remain separate preparations. Their common receive-transform-transmit form makes the routes comparable; it does not make their carriers or biological consequences identical.

16.2 The nested model and the refusal ladder

The model ladder is useful only if each addition answers a problem left open by the preceding model. M0 asks how much course correction follows the measured wound field and receiver state. M1 asks whether an immune relay adds information. M2 asks whether a measured neural or neuropeptide variable adds information. M3 asks whether a cross-fitted receiver-relative difference adds information after all of those variables are represented.

For changed gradient geometry, the inspectable model adds an explicit double-angle transform. If the normalized gradient vector is $\mathbf{g} = (g_x, g_y)$, the candidate transformed direction is

$$T_2(\mathbf{g}) = \frac{(g_x^2 - g_y^2,\ 2g_xg_y)}{\left\|(g_x^2 - g_y^2,\ 2g_xg_y)\right\| + \epsilon}. \tag{45}$$

This feature does not assert that living immune cells compute a double-angle map. It makes one declared synthetic failure family representable so the analysis can distinguish model misspecification from absence of every route.

The geometry-aware unit summary joins six groups:

$$\mathbf{x}_a = [\mathbf{s}_a, \Delta\mathbf{s}_a, \mathbf{d}_a, \mathbf{c}_a, \mathbf{z}_a, \mathbf{q}_a]\,, \tag{46}$$

where $\mathbf{s}_a$ contains the four frozen model scores, $\Delta\mathbf{s}_a$ their three ordered increments, $\mathbf{d}_a$ signed alignment summaries, $\mathbf{c}_a$ cross-product and transformed-geometry summaries, $\mathbf{z}_a$ receiver and motion summaries, and $\mathbf{q}_a$ field-dependence, retention, and measurement-reliability quantities. Every element is computed at the independent synthetic-chamber level.

16.3 The adverse robustness record remains governing

The origin-set development work does not replace the generator-shift robustness result. The preserved classifier passed three of ten non-mixture families, produced five adverse cases, and was correctly refused in two additional cases. Its 24-row adverse, null, and refusal ledger remains byte-identical. Its eleven Holm-retained null tests and all three mixture refusals remain part of the interpretation.

The confusion matrices expose the failure structure. Under a weak newest contribution, M1 collapsed into M0, M2 collapsed into M1, and M3 collapsed into M2. Under the double-angle field, M1 and most M2 units were absorbed into M0. These are not small calibration errors around otherwise stable route identity; they show that the frozen additive templates were not portable across those declared changes.

Calibration also moved with the generator. The matched family was both correct and confident. Weak coefficients produced confident errors. Altered geometry and nonlinear saturation produced distinct confidence-error patterns. A low expected calibration error on one generator therefore cannot become a general claim about uncertainty or biological confidence.

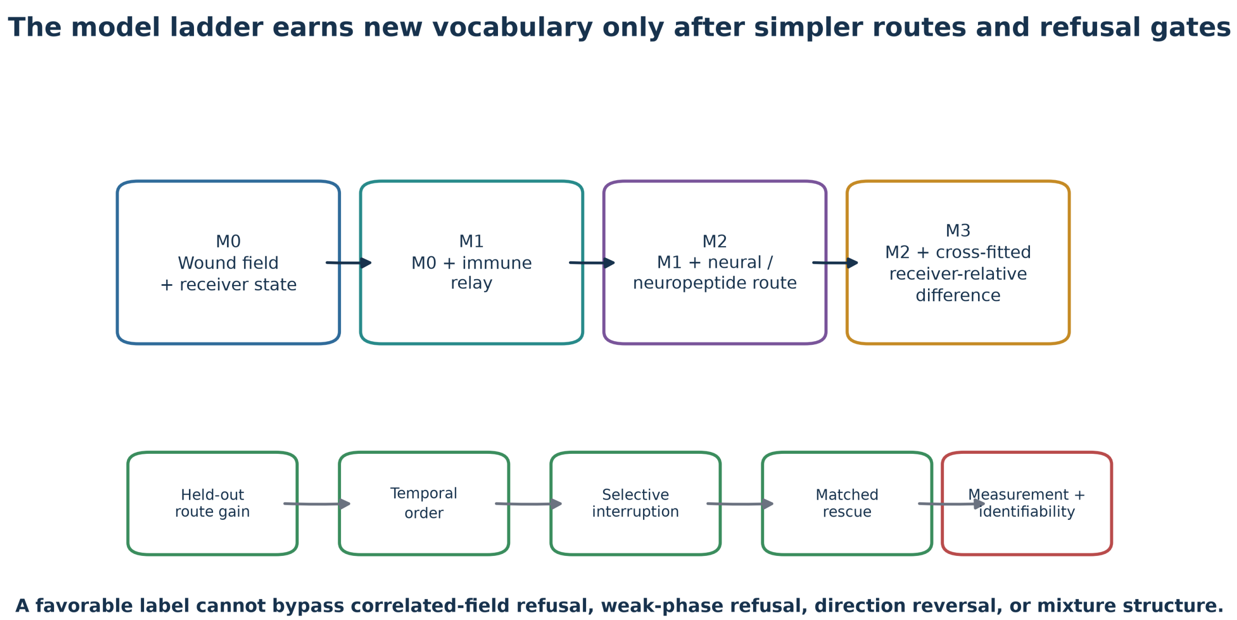

Figure 2: Nested model and refusal ladder. The predictive ladder begins with wound fields and receiver state (M0), then adds immune relay (M1), neural or neuropeptide input (M2), and a cross-fitted receiver-relative difference (M3). A new term earns a route interpretation only after held-out gain, temporal order, selective interruption, justified rescue, measurement reliability, and identifiability. A label cannot bypass a refusal gate.

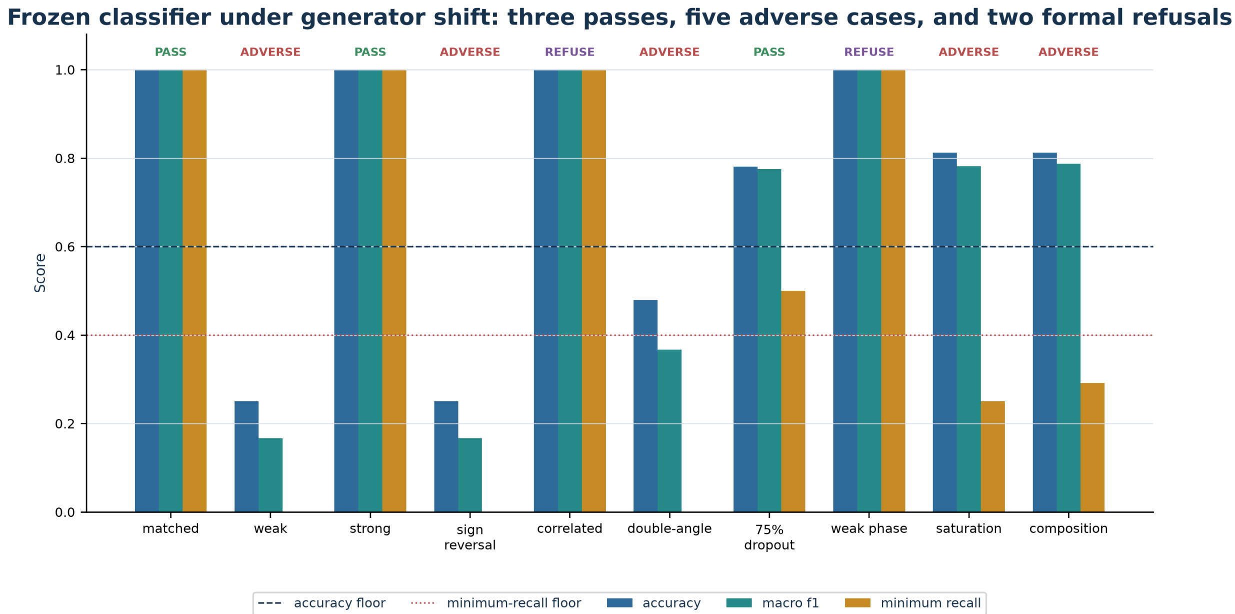

Figure 3: Generator-shift robustness dashboard. Accuracy, macro-F1, and minimum route recall for the preserved classifier across ten frozen non-mixture stress families. Three families passed the complete numerical and refusal gate, five were adverse, and correlated fields plus weak observed phase were formally refused despite perfect route labels. The displayed thresholds are only part of the complete gate, which also required macro-F1 and calibration conditions.

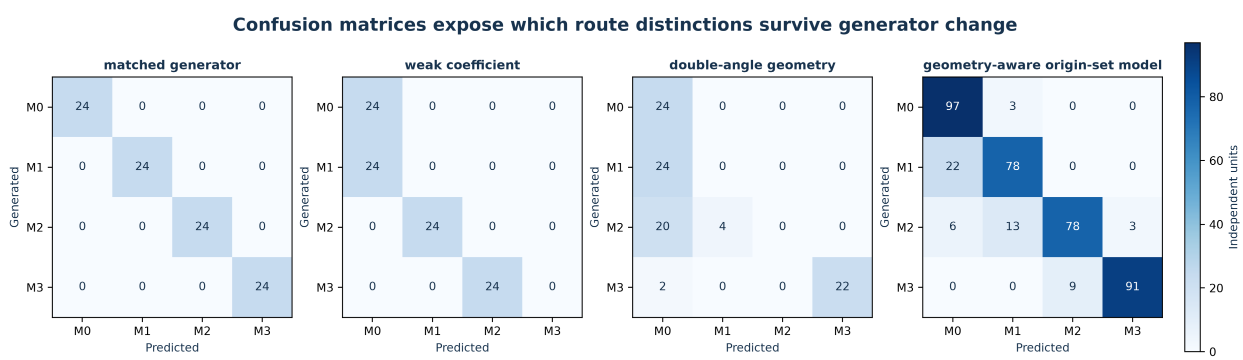

Figure 4: Confusion matrices under changed generators. Matched, weak-coefficient, and double-angle robustness matrices are contrasted with the geometry-aware internal-development result pooled across single-origin generator families. Each count is an independently generated synthetic chamber. The 400-unit pooled panel displays error structure and is not a directly comparable sample-size estimate for one 96-unit family.

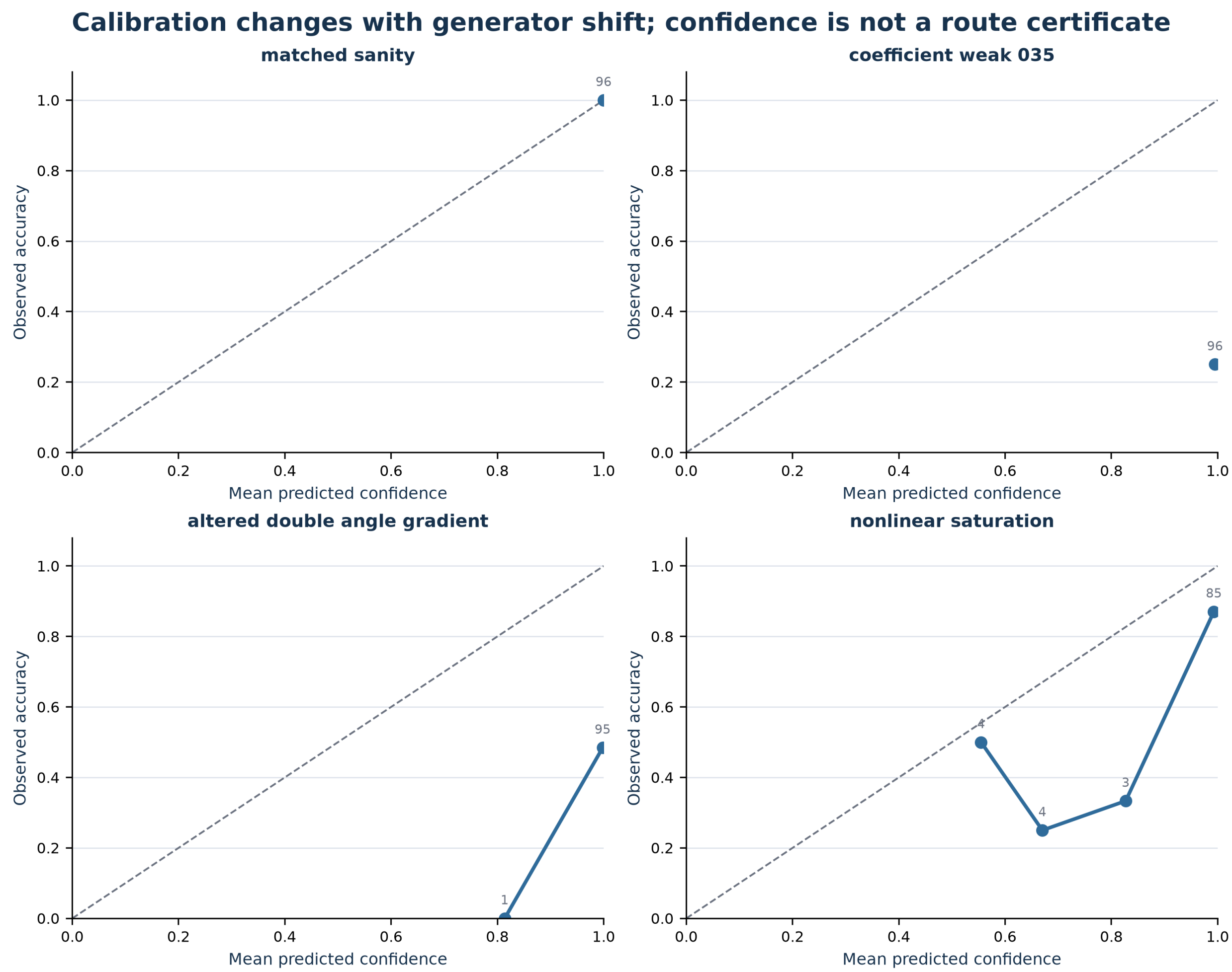

Figure 5: Calibration under generator shift. Reliability curves for four frozen generator families; point labels give independent-unit counts in populated confidence bins. Perfect matched confidence did not transfer to weak, altered-geometry, or nonlinear generators. Calibration is distribution-dependent and cannot certify a route by itself.

16.4 Two failed estimator repairs remain part of the record

The first development estimator used thirty unit-level summaries and a multi-label component model. On its independently seeded internal-development holdout, the geometry-aware route classifier reached 0.815 accuracy, 0.812 macro-F1, and 0.63 minimum route recall. The component model recovered 82.9 percent of mixture component sets exactly, but compressed one of seventy mixtures into a singleton. That 1/70 error failed the frozen zero-compression gate.

The first bounded repair did not retune the used holdout. It created a new partition and allowed calibration to select a bounded uncertainty guard around the secondary-component threshold. The repair eliminated mixture-to-singleton compression and raised exact mixture recovery to 92.9 percent. It also over-expanded many genuine singleton cases: singleton exactness fell to 60.5 percent, below the frozen 70-percent requirement. The apparent repair therefore moved the error instead of solving it.

Both outcomes matter. A system can avoid false singularity merely by labeling too many cases as mixtures. Conversely, a system can preserve singleton precision by compressing weak secondary origins. A valid route-origin estimator must measure both errors on data not used to choose the rule.

16.5 The origin-set estimator

The origin-set estimator again used a new development, calibration, and internal-holdout partition. It replaced independent component thresholds with ten mutually exclusive origin-set classes: four singleton routes and six unordered pairs. Because the development corpus contains many more singleton instances than pair instances, each class received inverse-frequency weight. For standardized feature matrix X, one-hot origin-set matrix Y, diagonal sample-weight matrix W, and ridge penalty λ, the fitted coefficient matrix is

$$\widehat{B}_\lambda = \left(X^\mathsf{T} W X + \lambda P\right)^{-1} X^\mathsf{T} W Y, \tag{47}$$

where the intercept is unpenalized. Ridge penalty and probability temperature were selected only on calibration units. The calibration objective first maximized the worse of singleton and mixture exactness, then overall exactness and mean Jaccard overlap. The internal holdout was used once after those choices.

For admissible origin set S_j, the reported development probability is

$$\widehat{p}(S_j \mid \mathbf{x}_a) = \frac{\exp\{(\mathbf{x}_a^\mathsf{T} \widehat{B})_j / T\}}{\sum_{\ell=1}^{10} \exp\{(\mathbf{x}_a^\mathsf{T} \widehat{B})_\ell / T\}}. \tag{48}$$

On 400 single-origin internal-development units, the geometry-aware route classifier achieved 0.860 accuracy, 0.860 macro-F1, and 0.78 minimum route recall. The M0, M1, M2, and M3 recalls were 0.97, 0.78, 0.78, and 0.91. Its ECE was 0.191. On the complete 470-unit origin-set holdout, the ten-class model achieved 0.800 exact set recovery and 0.821 mean Jaccard overlap. Singleton exactness was 0.825. Mixture exactness was 0.657, just above the frozen 0.65 floor. No mixture was compressed into a singleton.

The aggregate pass does not erase uneven families. Exact recovery was 0.50 for the 30:70 M1-M3 mixture, 0.60 for the equal M0-M1 and M1-M2 mixtures, 0.70 for three pair families, and 0.80 for equal M0-M3. Weak-coefficient, sign-reversal, and double-angle single-origin families also remained adverse in the geometry-aware route ledger. The origin-set estimator passed the development gate because all nine frozen aggregate, refusal, and interpretation conditions passed; it did not make every generator easy.

16.6 Interpretation requires both dependence and refusal checks

Feature ablation shows which summaries the fitted estimators used. In the generator-shift audit, removing the planted gradient, relay, neural, innovation, or phase group reduced matched-generator score in every tested unit. In the origin-set estimator, replacing each feature group by its development mean changed internal-holdout loss, with receiver-state summaries carrying the largest model dependence. These results diagnose the implementation. They do not establish that a corresponding feature is necessary in a living cell.

The refusal rule remains logically prior to a route certificate. Let G_{id} indicate adequate field independence, G_{meas} adequate measurement reliability, G_{dir} preservation of the claimed direction, and G_{mix} an admissible origin-set interpretation. Then

$$\text{Eligible}_a = G_{id,a} G_{meas,a} G_{dir,a} G_{mix,a} \nVdash [\text{predictive gate passes}] \nVdash [\text{selective causal gate passes}]. \tag{49}$$

The origin-set estimator correctly refused field attribution in every correlated-field internal-development unit and refused a phase or PWD certificate in every weak-observed-phase unit. A sign-reversal generator remained ineligible for a direction-preserving certificate. A synthetic origin-set label supplied no biological causal gate.

The trajectory panels make the generator changes visible without implying microscopy. Each line is the integrated displacement of one synthetic cell. Changed route strength, transformed gradient geometry, and a degraded observed phase channel alter the sampled paths even when some aggregate labels remain recoverable.

16.7 Development-estimator result and boundary

The development program passes the architecture, causal-figure, confusion, calibration, ablation, trajectory, development-estimator, interpretability, refusal, and deterministic-replay gates. The origin-set primary and replay scientific files are byte-identical. The result remains development-only. It does not open or evaluate a final confirmation partition, use living data, validate neural steering, establish a phase-bearing immune signal, validate PWD in an immune preparation, or support clinical action.

The decision logic and preservation invariants were therefore subjected to bounded formal verification. That layer proves facts about finite origin sets, refusal precedence, nested-route labels, and certificate composition in the stated abstract model. It cannot prove that the biological premises are true. The subsequent integrated review checks source reconstruction, medicine, statistics, nonduplication, executable replay, figure fidelity, and the boundary between machine theorem and biological evidence.

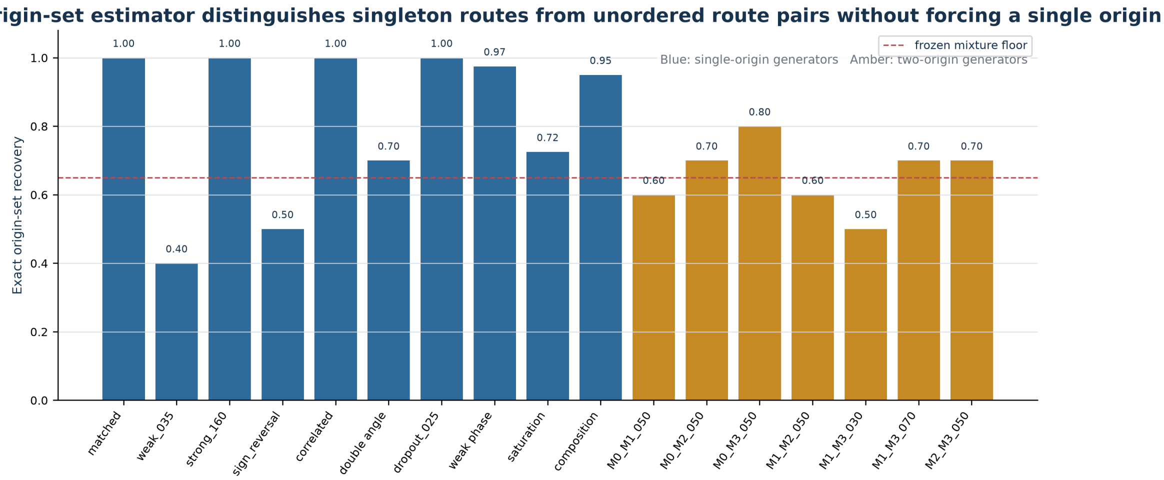

Figure 6: Origin-set recovery. Exact recovery by generator for the class-balanced ten-origin-set estimator. Blue bars are single-origin generators and amber bars are two-origin generators. The red line is the frozen aggregate mixture exactness floor, not a per-generator acceptance line. Aggregate mixture exactness was 0.657; no mixture was compressed into a singleton, while several individual pair families remained difficult.

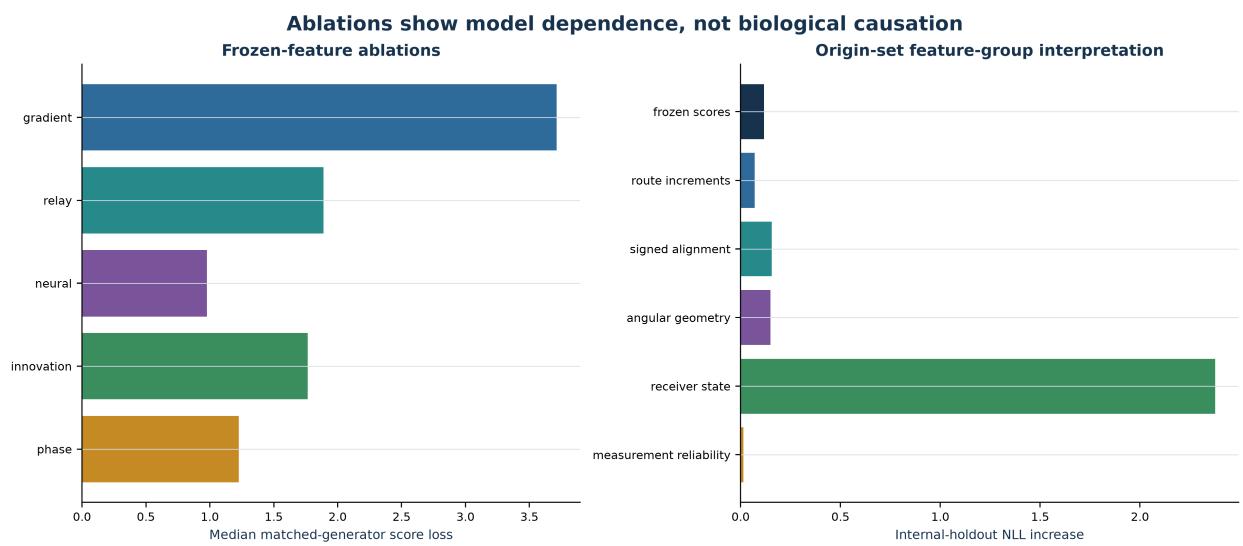

Figure 7: Feature ablation and interpretation. Left: median matched-generator score loss after frozen-coefficient feature ablation. Right: increase in internal-holdout negative log likelihood after replacing each origin-set feature group with its development mean. These are fitted-model dependence tests, not evidence that the corresponding biological route exists or is causal.

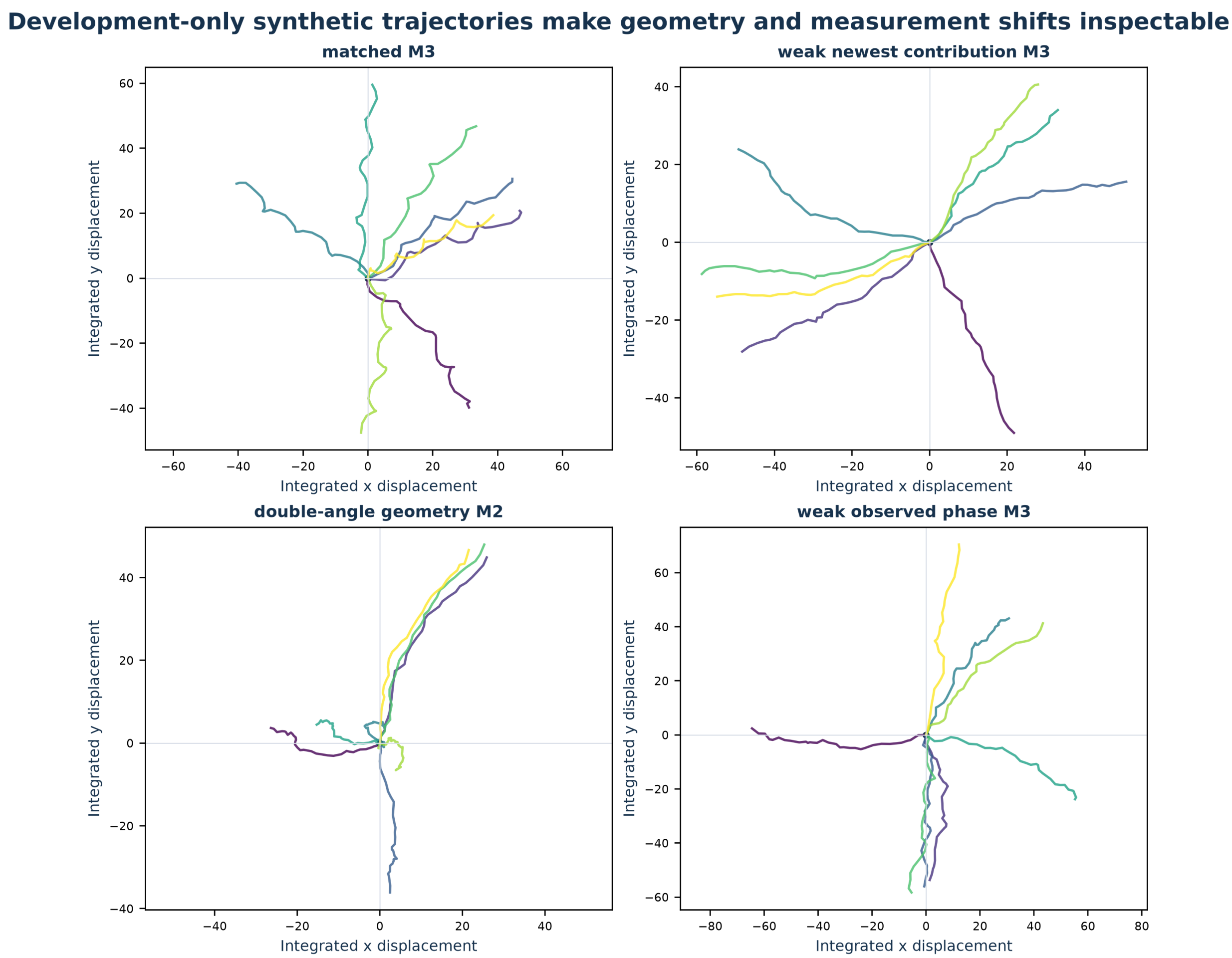

Figure 8: Development-only trajectory examples. Integrated two-dimensional displacements for eight cells in one independently seeded internal-development unit from four declared generator families. The panels make changes in route strength, gradient geometry, and observed phase reliability inspectable. They are synthetic trajectories, not microscopy, animal, or human data.

17 Machine-checking what a route certificate is allowed to mean

17.1 The problem comes before the proof vocabulary

A route classifier can produce a confident label even when the variables needed to interpret that label are not independently measurable. A phase-associated score can be positive when the phase channel is unreliable. A route label can be correct when two candidate fields are nearly collinear. A mixed generator can be forced into one label even though two origins contributed. These are not merely errors in accuracy. They are errors in what the output is allowed to mean.

The scientific account already requires separate evidence for measurement, prediction, causal selectivity, direction, and route identity. The formal layer asks a narrower mathematical question: once those requirements are encoded as finite logical gates, do the refusal and compatibility rules actually follow? The proof cannot determine whether a living preparation satisfies a premise. It can determine whether the stated decision system accidentally issues a certificate after a premise has failed.

Only after that problem is visible do the formal names help. The Lean model uses four ordered route labels, ten admissible singleton-or-pair origin sets, six Boolean certificate gates, and typed compatibility edges. The origin-set universe is

$$\mathcal{O} = \{M_0, M_1, M_2, M_3, M_0{+}M_1, M_0{+}M_2, M_0{+}M_3, M_1{+}M_2, M_1{+}M_3, M_2{+}M_3\}. \tag{50}$$

This is a bounded ontology for the development experiment, not a claim that biological routes are limited to these ten possibilities. Triples, higher-order combinations, unmodeled routes, and unknown origins remain outside the type.

17.2 Refusal is part of the result

Let the encoded gates denote field independence, measurement reliability, direction preservation, origin admissibility, held-out predictive gain, and selective causal support. Eligibility is their conjunction:

$$E(g) = G_{id} G_{meas} G_{dir} G_{origin} G_{pred} G_{causal}. \tag{51}$$

Lean machine-checks six refusal theorems: setting any one gate to false makes $E(g)$ false. It separately checks that an eligible claim must have a reliable measurement gate and that a phase certificate therefore cannot survive measurement failure. This formalizes the same boundary exercised by the weak-observed-phase development family: a route label may remain recoverable while the phase/PWD branch is unavailable.

The model also supplies a concrete counterexample to the inference from label to certificate. Its candidate has a positive M3 label and all gates except field independence, yet the certificate is refused. The theorem does not estimate how often such a case occurs. It establishes that positive classification and admissible causal interpretation are distinct types of conclusion.

17.3 Mixtures and complete paths

The ten origin-set constructors distinguish four singletons from six pairs. Lean checks that every encoded mixture is non-singleton. That finite theorem is the logical counterpart of the zero mixture-to-singleton gate;

it does not establish that the origin-set estimator always identifies the correct pair.

For an ordered list of typed edges $e_1, \ldots, e_k$, the encoded path certificate is

$$C(e_{1:k}) = \prod_{j=1}^{k} \nVdash[e_j \text{ is present}]\nVdash[r(\text{target}\, e_j) = r(\text{source}\, e_j) + 1]. \tag{52}$$

The compiler verifies that any listed edge whose presence flag is false forces $C = 0$, and that the declared $M_0 \to M_1 \to M_2 \to M_3$ chain is adjacent in the encoded rank. The implication is deliberately one-way:

$$\begin{aligned} C(e_{1:k}) = 1 &\implies \text{the declared finite path is internally compatible,} \\ C(e_{1:k}) = 1 &\not\Rightarrow \text{the biological edges exist.} \end{aligned} \tag{53}$$

17.4 Compiler and axiom audit

The exact file `MIBBRouteLogic.lean` passed the shared Lean compiler lease with exit code 0. It contains fourteen named theorems and no `sorry`, admitted proof, or external biological axiom. Four concrete enumeration and accounting theorems report no axioms. Ten general proofs that use Lean simplification report only Lean's standard proposition-extensionality axiom, `propext`. The machine check therefore closes the stated finite logic under the ordinary Lean foundation; it does not promote any empirical premise.

The typed claim signature and five compatibility edges carry the model, requirements, scope, and forbidden overclaims. The formal result is strongest where it should be: it prevents a later implementation or prose revision from turning a failed measurement, correlated field, missing edge, or mixed origin into a complete certificate by definition. It remains silent about whether living neural and immune systems instantiate those objects.

17.5 Formal result and boundary

The finite proof suite passes the bounded compiler, theorem-shape, typed-signature, compatibility-edge, refusal-precedence, mixture-disjointness, and count-preservation gates. The earlier development artifacts remain hash-preserved. The development estimator and all adverse results are unchanged. No final-confirmation partition or biological dataset was used.

The integrated review checks strongest-form source reconstruction, medical and statistical review, nonduplication, executable-evidence replay, figure readback, and a line-by-line check that the machine theorem is not narrated as biological proof. The release-production stage follows only after those boundaries pass.

18 Route completeness, uncertainty, and strongest-form review

18.1 A body–brain network includes interfaces, not just end points

The central mechanism is easiest to miss when the brain and immune system are drawn as two boxes connected by one arrow. A peripheral signal can be transformed at a sensory ending, ganglion, brainstem nucleus, autonomic efferent, vascular barrier, meningeal interface, local glial population, or immune-cell receptor. The consequence that reaches a neuron need not be the original molecule that left the tissue. Conversely, a neural consequence can change endocrine output, autonomic tone, vascular state, immune-cell retention, or local cytokine production without directly contacting a moving leukocyte.

Brain borders are active parts of this architecture. Meningeal lymphatic vessels carry fluid and immune cells from cerebrospinal fluid toward deep cervical lymph nodes [40]. Dural sinuses accumulate CNS-derived antigens that can be captured by antigen-presenting cells and shown to patrolling T cells [41]. Skull and vertebral marrow can supply myeloid cells to meninges and, in defined pathological preparations, to CNS parenchyma [42]. Skull marrow also supports a meningeal B-cell niche [43], and cerebrospinal fluid can enter skull marrow through dural channels and alter hematopoietic activity after injury [44]. These routes support bidirectional communication while preserving a necessary ceiling: participation at meninges, dural sinuses, marrow interfaces, or circumventricular organs is not evidence that immune cells routinely cross healthy parenchyma or contact every neuron.

The vascular interface is likewise a receiver and transformer rather than a passive open-or-closed wall. Endothelial cells, pericytes, and perivascular macrophages respond to peripheral immune challenges. Competing cytokine programs can stabilize or weaken barrier integrity [45], and inflammatory challenge can reprogram distinct barrier-cell populations toward prostaglandin signaling [46]. A complete route certificate must therefore distinguish trans-barrier passage of a molecule from a barrier-mediated downstream transformation.

18.2 Alternative route families must compete

For a named endpoint, at least six explanation classes should remain available:

1. **Generic-load model:** total inflammatory or stress burden predicts the endpoint without a typed route.
2. **Humoral/endothelial model:** circulating mediators and vascular or barrier transformations explain the effect.
3. **Autonomic/endocrine model:** neural or hormonal state predicts the endpoint without the proposed mobile-cell edge.
4. **Border-surveillance model:** meningeal, lymphatic, marrow, or perivascular interfaces explain the effect without a complete COT route.
5. **Typed feedforward route:** ordered carriers and receivers improve held-out prediction.
6. **Recurrent typed network:** fast and slow states, returned physiology, and cross-route interactions improve prediction and next-step rescue.

The distinctive proposal begins only when the typed models outperform the simpler alternatives under experimental-unit-held-out scoring and selective intervention. If a generic, humoral, autonomic, endocrine, endothelial, or border model predicts equally well with fewer degrees of freedom, the more elaborate route claim should be rejected.

18.3 The vagus–spleen dispute is an identifiability test

The inflammatory reflex is often compressed into a vagus-to-celiac-to-splenic-nerve-to-T-cell chain. Experiments support a splenic catecholaminergic component and acetylcholine-producing T cells in the

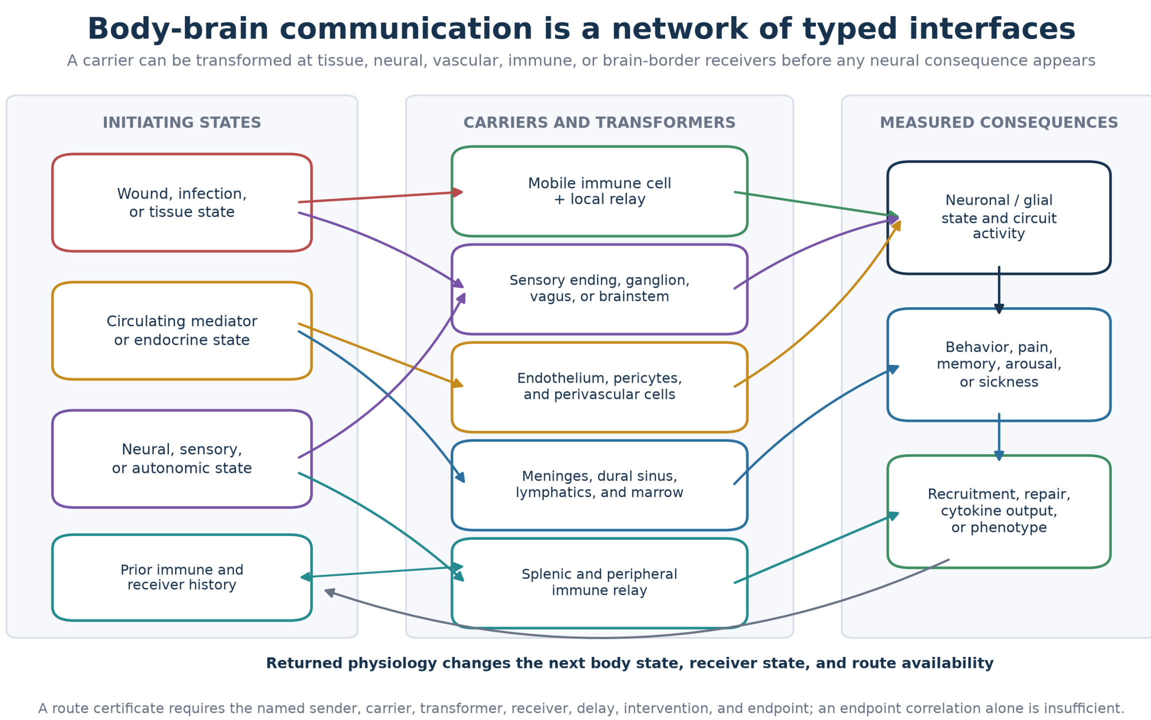

Figure 9: Route completeness across body-brain interfaces. Peripheral, humoral, neural, immune, vascular, meningeal, lymphatic, marrow, and splenic transformations remain separate before the measured neural, behavioral, or immune consequence. Returned physiology changes the next body and receiver state. A route certificate requires the named sender, carrier, transformer, receiver, delay, intervention, and endpoint; endpoint correlation alone is insufficient.

downstream transformation [5,34]. Electrophysiological and anatomical analyses have disputed a direct vagal drive of splenic sympathetic neurons [35,36]. This is not a reason to discard neuroimmune communication or to preserve a disputed arrow. It is a worked example of typed correction: the downstream transformations may remain supported while the upstream route remains an alternative set.

```
vagus intervention
|-- proposed celiac or splenic neural relay
|-- central autonomic coordination with sympathetic output
|-- endocrine or humoral co-route
`-- preparation-dependent mixed route
        |
        v
splenic catecholamine-sensitive immune state
-> ChAT-positive T-cell acetylcholine
-> macrophage alpha-7 nicotinic receptor
-> TNF consequence
```

Older vagotomy experiments also support a vagal contribution to IL-1-associated temperature and sickness physiology [37,38], while residual noradrenergic and hypothalamic–pituitary–adrenal responses after vagotomy show that the vagus is not the only immune-to-brain path [39]. The correct scientific question is therefore which edge carries which consequence in a named preparation, not whether one iconic diagram is universally true.

18.4 Statistical uncertainty around the development result

The frozen development gate reported deterministic point estimates because its purpose was a frozen development gate. The integrated review adds uncertainty without refitting any estimator or changing any threshold. Wilson intervals treat each synthetic chamber as the independent unit. Stratified bootstrap intervals resample chambers within true route or generator family. These are uncertainty summaries for the declared development distribution, not estimates of biological performance.

The geometry-aware classifier recovered 344 of 400 single-origin units: accuracy 0.860, with a 95-percent Wilson interval of 0.823–0.891. Its stratified-bootstrap macro-F1 interval was 0.825–0.892 around the observed 0.860. Route-recall intervals were 0.915–0.990 for M0, 0.689–0.850 for M1, 0.689–0.850 for M2, and 0.838–0.952 for M3.

The origin-set estimator recovered 376 of 470 sets exactly: 0.800, with a 95-percent Wilson interval of 0.761–0.834. Singleton exactness was 330/400, 0.825 (0.785–0.859). Mixture exactness was 46/70, 0.657 (0.540–0.758). Mean Jaccard overlap was 0.821, with a stratified-bootstrap interval of 0.793–0.849. No mixture was compressed into a singleton in 70 development cases, but the corresponding Wilson lower bound is 0.948, not certainty about future cases. Likewise, 40/40 correlated-field refusals and 40/40 weak-phase refusals each have a Wilson lower bound of 0.912.

The family-level mixture intervals are necessarily wide because each pair has only ten internal-holdout units. The 30:70 M1–M3 case recovered 5/10, whose Wilson interval is approximately 0.237–0.763. That uncertainty prevents the aggregate 0.657 mixture result from being narrated as uniformly stable pair recovery.

18.5 Reproducibility, figure readback, and formal-boundary review

A fresh review replay regenerated all twelve R2 scientific files byte-for-byte, including predictions, model record, adverse ledger, inherited generator-shift ledger, gate, and manifest. It again passed all nine development gates. This is executable reproducibility on the same declared software environment, not external replication.

All eight figures were read against their bound data and captions. The architecture figure distinguishes carriers, receivers, transformations, and return paths. The causal ladder places refusal before interpretation. The failure dashboard, confusion matrices, and calibration panels retain adverse generator shifts. The ablation panel is labeled as fitted-model dependence rather than biological necessity. The origin-set panel shows uneven pair recovery. The trajectory panel is explicitly synthetic and is not presented as microscopy.

The formal prose also passed a claim-boundary readback. Every theorem is described as a consequence of the finite encoding. No compiler result is used to establish a living carrier, receiver, immune-cell phase channel, PWD, clinical benefit, or causal route. The typed proof strengthens the refusal discipline without replacing medicine or experiment.

18.6 Scope relative to companion papers

This paper owns the typed neuroimmune route graph, wound-navigation hierarchy, receiver-relative residual test, staged perturbation and rescue program, generator-shift application, origin-set development estimator, and formal refusal logic. *Super Evolution Theory* owns the broader receptor ecosystem, signal lifecycle, structural adaptation, natural selection, and horizontal-transfer argument. *From Tripartite Synapses to Multicellular Engrams* owns neuron–astrocyte roles in memory content, access, stabilization, and recall. *From Gamma Synchrony to Informative Difference* and the broader PWD-calculus program own the general information-theoretic and oscillatory treatment. This paper imports only the bounded receiver-relative test needed for immune navigation.

The integrated architecture keeps brain-border routes, barrier transformations, the vagus–spleen dispute, competing route families, and the foundational 2017–2022 genealogy in the same account as wound navigation, experimental-unit inference, robustness, mixture recovery, figures, and formal logic. This release is the canonical current version of that unified research program, not a separate paper created from one subset of the routes.

18.7 Integrated-review and release boundary

The source-faithful, medical, scale, statistical, replay, figure, formal-boundary, and nonduplication reviews pass with explicit limits. They add no final-confirmation outcome and do not change the frozen estimator. The preprint therefore reports development evidence only. A future confirmatory partition requires a new blind commitment created by a separate operator; no confirmatory data were generated or opened during release production. All mechanisms, adverse results, refusals, uncertainty intervals, historical dates, route disputes, source boundaries, and formal limits remain in the manuscript or its hash-bound companion archive.

19 Scope and conclusion

The body and brain coordinate immunity through multiple physical routes: wound fields, mobile-cell relays, neuropeptides, cytokines, vagal sensory codes, brainstem control, meningeal and parenchymal immune populations, glia, neuronal receptors, and behavior. Treating these routes as one vague axis conceals the mechanisms that experiments can actually decide.

The proposed architecture keeps the routes typed and joins them through recurrent state change. It then isolates one additional SAN question: whether a moving immune receiver uses a timing- or spectral-relative residual that carries directional information beyond established gradients and relays. The paper supplies the equations, nested baselines, perturbations, rescues, synthetic benchmark, and falsifiers needed to answer that question. The result can therefore improve either way: a positive result earns a new receiver-relative mechanism; a negative result locates neural influence in gain, state, or feedback rather than steering.

The executed application sharpens that conclusion. It can recover some declared synthetic route structures, preserve refusal under named attribution failures, expose generator-shift fragility, and represent mixtures without forcing a singleton label. Its uneven pair recovery and adverse families are not secondary caveats; they define where the present estimator is not yet reliable. The formal proof adds a different guarantee: once measurement, prediction, selectivity, direction, and compatibility premises are encoded, the finite certificate logic cannot legitimately bypass a failed premise. Neither result substitutes for living-data intervention and rescue.

The research program is therefore experimentally asymmetric in a productive way. If established gradients, relays, neural signals, and receiver history fully explain held-out course correction, the phase or spectral extension should be rejected. If a reliable receiver-relative timing term adds information, survives matched controls, disappears under the named carrier or receiver perturbation, and returns under a next-step rescue, it earns a bounded new mechanism. Either result improves the route map.

Reproducibility, data, and ethics

The public-safe companion package contains the frozen synthetic contracts, source and claim ledgers, application code, complete development outputs, replay receipts, statistical readback, figures, Lean source, compiler and axiom receipts, adverse and refusal ledgers, and validation manifests. The development application is synthetic; no new biological or clinical data are reported. The biological protocols are prospective and require institutional animal-care review, local biosafety review, trained personnel, preregistered experimental-unit analysis, and an independently justified sample size before execution. The staged program prioritizes existing data, in vitro and ex vivo calibration, and identifiable experiments before animal expansion.

The source and genealogy ledgers distinguish dated SAN formulations from later scientific evidence and from the present synthesis. The same separation applies to software evidence: deterministic replay and machine-checked finite logic are reproducibility results, not biological confirmation.

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